

High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number:
Submittal Title: For Information Only
TO:
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec:	Paragraph:
Authored By:	Date Submitted:

Submittal Certification	
Check Either (A) or (B):	
☐ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title:	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED _____
SPECIFICATION SECTION # _____
SPECIFICATION _____
PARAGRAPH _____
DRAWING _____
SUBCONTRACTOR _____
SUPPLIER _____
MANUFACTURER _____

CERTIFIED BY CQCM or Designee : 

Submittal



SUB Reference No : MECI-SUB-0007

Status: OUTSTANDING

For Action: Darrin Lee, MECI
David Wieseler, MECI

Project: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT

Subject: 16060 - GROUNDING AND BONDING

Submittal for grounding and bonding, including compression connectors, ground rods, ground cables, precast ground well boxes, grounding electrode conductors, and bonding jumpers.

Submitted For: Approval

Specification Reference : 16060 - GROUNDING AND BONDING

Paragraph No.: 16060 - 1.03.A

Description : Submittal for grounding and bonding, including compression connectors, ground rods, ground cables, precast ground well boxes, grounding electrode conductors, and bonding jumpers.

Discipline: ELEC

Area: 00

Attachments: 16130 - CONDUITS - GROUNDING AND BONDING.pdf

SUB by:	Hannah Anderson, MECI	On: 09 September 2025
Created By:	Hannah Anderson, MECI	On: 09 September 2025, 01:09:20 PM -10:00
Last Edited By:	Hannah Anderson, MECI	On: 09 September 2025, 01:44:44 PM -10:00

01 - QC/Project Engineer Review

APPROVED WITHOUT COMMENT.

QC/Project Engineer Review by:	David Wieseler, MECI	On: 09 September 2025
Created By:	David Wieseler, MECI	On: 09 September 2025, 01:46:10 PM -10:00
Last Edited By:	David Wieseler, MECI	On: 09 September 2025, 01:46:10 PM -10:00

02 - Project Manager Review

Project Manager Review
by:

On:

Created By:On: ,

Last Edited By:On: ,

03 - Response

Response by:On:

Created By:On: ,

Last Edited By:On: ,

04 - Project Manager/Engineer Closeout

Project Manager/Engineer
Closeout by:On:

Created By:On: ,

Last Edited By:On: ,

MASS ELECTRIC CONSTRUCTION COMPANY PRODUCT DATA SUBMITTAL

REVISION	DATE	PREPARED BY	APPROVED BY
0	09/09/2025	Hannah Anderson	David Wieseler

HILO WASTEWATER TREATMENT PLANT REHABILITATION AND REPLACEMENT PROJECT



SPECIFICATION - 16060 GROUNDING AND BONDING

DOCUMENT REVISION LOG						
Revision Number	Revision Date	Description	Approvals			
			PE INITIAL		PM INITIAL	
0	09/09/2025	Issued for Review	DCW	09/09/2025		

COMMENTS AND CLARIFICATIONS

- **Ground Rods (Spec 2.02.B)**
Per response to RFI #0039, the alternative of a pointed ground rod in lieu of a Burndy-type ground rod was approved. The product included in this submittal reflects the approved alternate.
- **Ground Cable (Spec 2.02.C)**
This submittal includes #6 and #8 stranded ground cable in place of the solid ground cable specified in 2.03.B.2.a. The substitution is due to the lack of manufacturer availability of solid conductors in these sizes.
- **Precast Ground Well Boxes (Spec 2.02.D)**
Precast ground well boxes will be provided with the required electrical stamp in accordance with specification 2.04.A.3.

SPECIFICATION – 16060 GROUNDING AND BONDING



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SECTION 16060
GROUNDING AND BONDING

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes:
1. Grounding materials and requirements.

1.02 REFERENCES

- A. Standards:
1. ASTM International (ASTM):
 - a. B8 - Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft.
 2. Institute of Electrical and Electronics Engineers (IEEE):
 - a. 81 - IEEE Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Grounding System.
 3. National Electrical Code (NEC).
 4. Underwriters Laboratories, Inc. (UL):
 - a. 467 - Ground and Bonding Equipment.

1.03 SUBMITTALS

- A. Furnish submittals as specified in Section 01330 - Submittal Procedures.
- B. Product data:
1. Catalog cutsheets.

1.04 QUALITY ASSURANCE

- A. All grounding components and materials shall be UL listed and labeled.

1.05 PROJECT OR SITE CONDITIONS

- A. As specified in Section 01850 - Design Criteria.

1.06 WARRANTY

- A. As specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

2.01 DESIGN AND PERFORMANCE CRITERIA

- A. Ground equipment and raceway systems so that the completed installation conforms to all applicable code requirements.

- B. Provide a complete electrical grounding system as indicated on the Drawings and as specified including but not limited to:
1. Grounding electrodes.
 2. Bonding jumpers.
 3. Ground connections.

2.02 MANUFACTURERS

- A. Compression connectors: One of the following or equal:

1. Burndy.
2. Erico.
3. Harger.
4. Panduit.
5. Thomas & Betts.

- B. Ground rods: One of the following or equal:

1. Erico.
2. Harger.
3. Nehring.
4. Thomas & Betts.

- C. Ground cable: One of the following or equal:

1. Erico.
2. Harger.
3. Nehring.
4. Southwire.

- D. Precast ground well boxes: One of the following or equal:

1. Brooks Products, 3-RT Valve Box.
2. Christy Concrete Products, G12 Valve Box.

2.03 MATERIALS

- ✓ A. Ground rod:
- ✓ 1. Minimum: 3/4-inch diameter, 10 feet long.
 - ✓ 2. Uniform 10 mil covering of electrolytic copper metallurgically bonded to a rigid steel core:
 - a. The copper-to-steel bond shall be corrosion resistant.
 - ✓ 3. In accordance with UL 467.
 - ~~4. Sectional type joined by threaded copper alloy couplings.~~
 - ~~5. Fit the top of the rod with a threaded coupling and steel-driving stud.~~
- B. Ground cable:
- ✓ 1. Requirements:
 - ✓ a. Soft drawn (annealed).
 - ✓ b. Concentric lay, coarse stranded in accordance with ASTM B8.
 - ✓ 2. Size is as indicated on the Drawings, but not less than required by the NEC. Use the following type of wire unless otherwise indicated on the Drawings:
 - ✓ a. Size #6 AWG and smaller; solid conductor.
 - ✓ b. Size #4 AWG and larger; stranded conductor.

- ✓ C. Compression connectors:
 - ✓ 1. Manufactured of high copper alloy specifically for the particular grounding application.
 - ✓ 2. Suitable for direct burial in earth and concrete.
 - ✓ 3. Identifying compression die number inscription to be impressed on compression fitting.
 - ✓ 4. Barrels prefilled and sealed with oxide-inhibiting and anti-seizing compound.
- D. Grounding electrode conductors:
 - 1. Minimum size in accordance with the NEC.
- ✓ E. Main bonding jumpers and bonding jumpers:
 - ✓ 1. Minimum size in accordance with the NEC.

2.04 ACCESSORIES

- ✓ A. Precast ground well boxes:
 - ✓ 1. Minimum 10-inch interior diameter.
 - ✓ 2. Traffic-rated cast iron cover.
 - ✓ 3. Permanent "GROUND" marking on cover.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Provide bonding jumpers and wire, grounding bushings, clamps and appurtenances required for complete grounding system to bond equipment and raceways to equipment grounding conductors.
- B. Provide a separate grounding conductor for each motor and connect at motor terminal box. Do not use bolts securing motor box to frame or cover for grounding connectors:
 - 1. When grounding motors driven by variable frequency drives (VFD) comply with the requirements of the VFD manufacturer.
- C. Provide a grounding type bushing with lug for connection of grounding conductor for conduits that originate from each motor control center section, switchboard, or panelboard:
 - 1. Individually bond these raceways to the ground bus in the equipment.
- D. Provide grounding type bushings with lugs for connection of grounding conductor at both ends of metallic conduit runs. Bond ground bushings to the grounding system.
- E. Provide a green insulated wire-grounding jumper from the ground screw to a box grounding screw and, for grounding type devices, to equipment grounding conductor.
- F. Interconnect the secondary switchgear, switchboard, or panelboard neutral bus to the ground bus in the secondary switchgear, switchboard, or panelboard compartment, only at service entrance point or after a transformer.

- G. Duct bank ground system:
 - 1. Provide a bare copper grounding conductor the entire length of each duct bank, embedded in the concrete of the duct bank as indicated on the Drawings and specified in the Specifications.
 - 2. Bond duct bank ground conductors together where duct banks join, merge, intersect, or split.
- H. Grounding at service (600 V or Less):
 - 1. Connect the neutral to ground only at one point within the enclosure of the first disconnecting means on the load side of the service transformer.
- I. Ground connections:
 - 1. All connections to the ground grid system, the duct bank grounding system, equipment, ground rods, etc., shall be made using compression type grounding connectors as indicated on the Drawings, UL listed, and labeled for the application.
 - 2. Make ground connections in accordance with the manufacturer's instructions.
 - 3. Do not conceal or cover any ground connections until the Engineer or authorized representative has established and provided written confirmation that every grounding connection is as indicated on the Drawings and specified in the Specifications.
- J. Grounding electrode system:
 - 1. Ground ring:
 - a. Provide all trenching and materials necessary to install the ground ring as indicated on the Drawings.
 - b. Ground ring conductor shall be in direct contact with the earth, or where embedded, concrete, of the size as indicated on the Drawings.
 - c. Minimum burial depth 36 inches or as indicated on the Drawings.
 - d. Re-compact disturbed soils to original density in 6-inch lifts.
 - 2. Ground rods:
 - a. Locations as indicated on the Drawings.
 - b. Length of rods forming an individual ground array shall be equal in length.
 - c. Drive ground rods and install grounding conductors before construction of concrete slabs and duct banks.
 - d. Pre-crimp all ground rods, as recommended by the manufacturer, before crimping connector to ground rod.
 - 3. Metal underground water pipe:
 - a. Bond metal underground domestic water pipe to grounding electrode system.
 - 4. Metal frame of building or structure:
 - a. Bond metal frame of building or structure to grounding electrode system.
 - 5. Extend grounding conductors through concrete to accessible points for grounding equipment and electrical enclosures.
 - 6. Where grounding conductors are exposed and subject to physical damage, install in Schedule 80 PVC conduit for protection.
 - 7. Install grounding system at each structure where switchgear, motor control centers, switchboards, panelboards, panels, or other electrical equipment are installed.

- K. Shield grounding:
 - 1. Analog signal cables shields shall only be grounded at a single point in the loop. Unless otherwise noted, ground signal cable shields at control panel.
 - 2. For communication and data line signal cable shields and drain wires should be grounded at both ends of the cable run.
 - 3. Insulate the shielding and exposed drain wire for each signal cable with heat-shrink tubing.
 - 4. Terminate the signal cable shield on a dedicated grounding terminal block.
- L. Where indicated on the Drawings, install ground rods in precast ground wells.

3.02 COMMISSIONING

- A. As specified in Section 01756 - Commissioning.
- B. Functional testing:
 - 1. As specified in Section 16950 - Field Electrical Acceptance Tests.

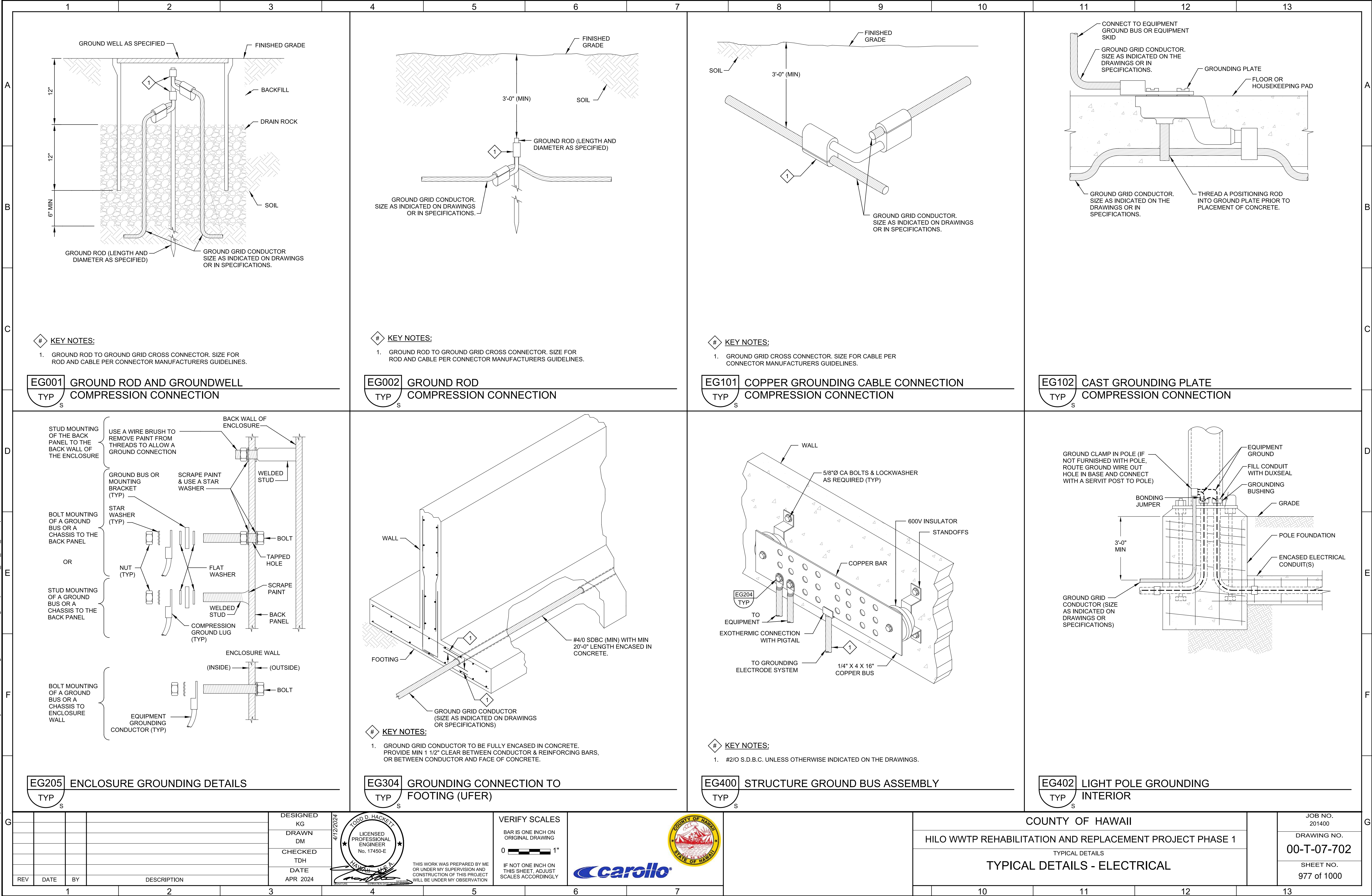
3.03 FIELD QUALITY CONTROL

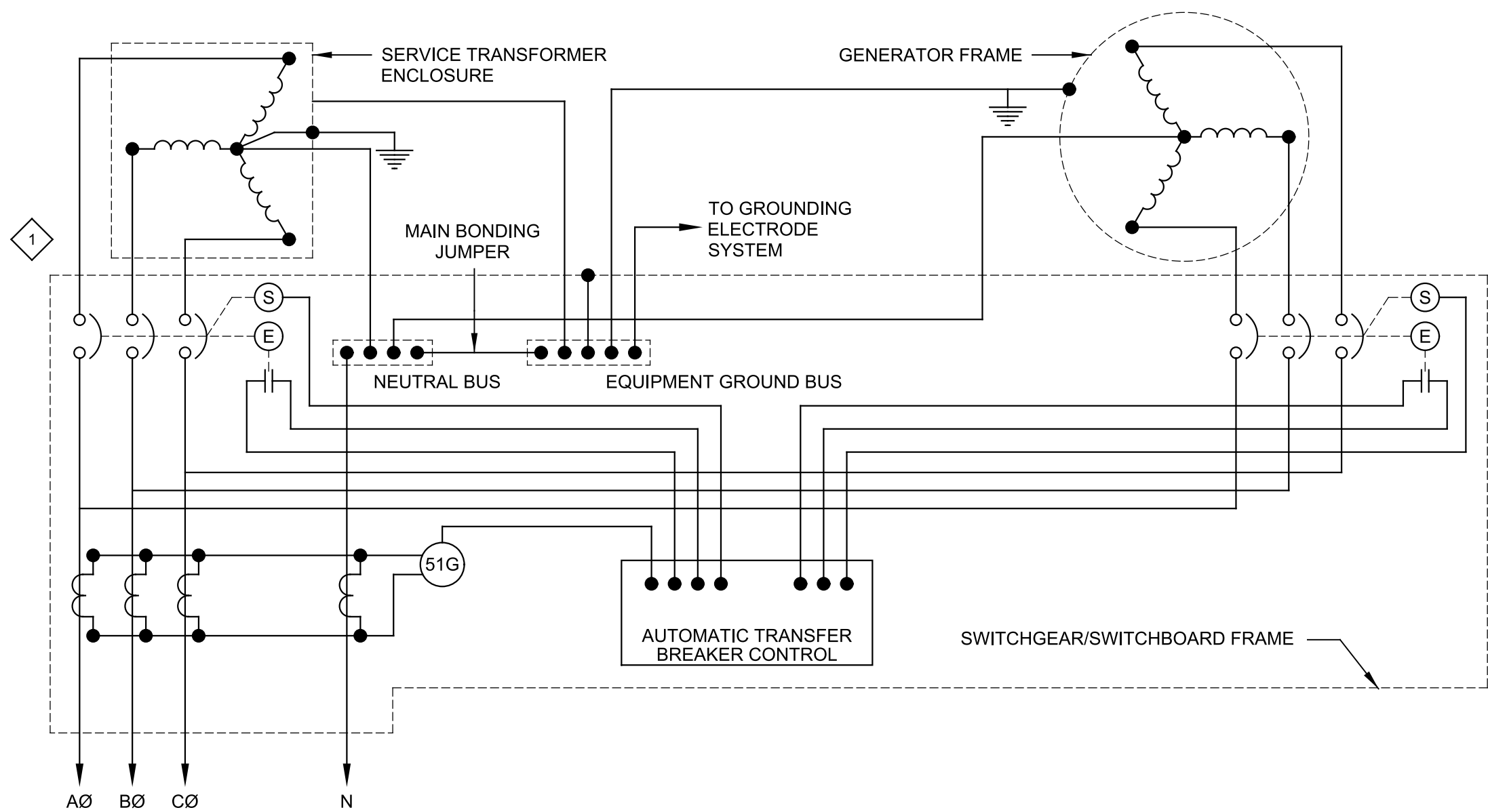
- A. The ground system resistance (electrode to ground) of the completed installation, as determined by tests, shall be:
 - 1. 5 ohms or less for industrial systems.
 - 1 ohm or less for electrical buildings.
- B. Measure grounding electrode system resistance to ground in accordance with IEEE 81.

3.04 ADJUSTING

- A. Under the direction of the Engineer, add additional parallel connected ground rods and/or deeper driven rods until the ground resistance measurement meets the specified resistance requirements:
 - 1. Use of salts, water, or compounds to attain the specified ground resistance is not acceptable.

END OF SECTION





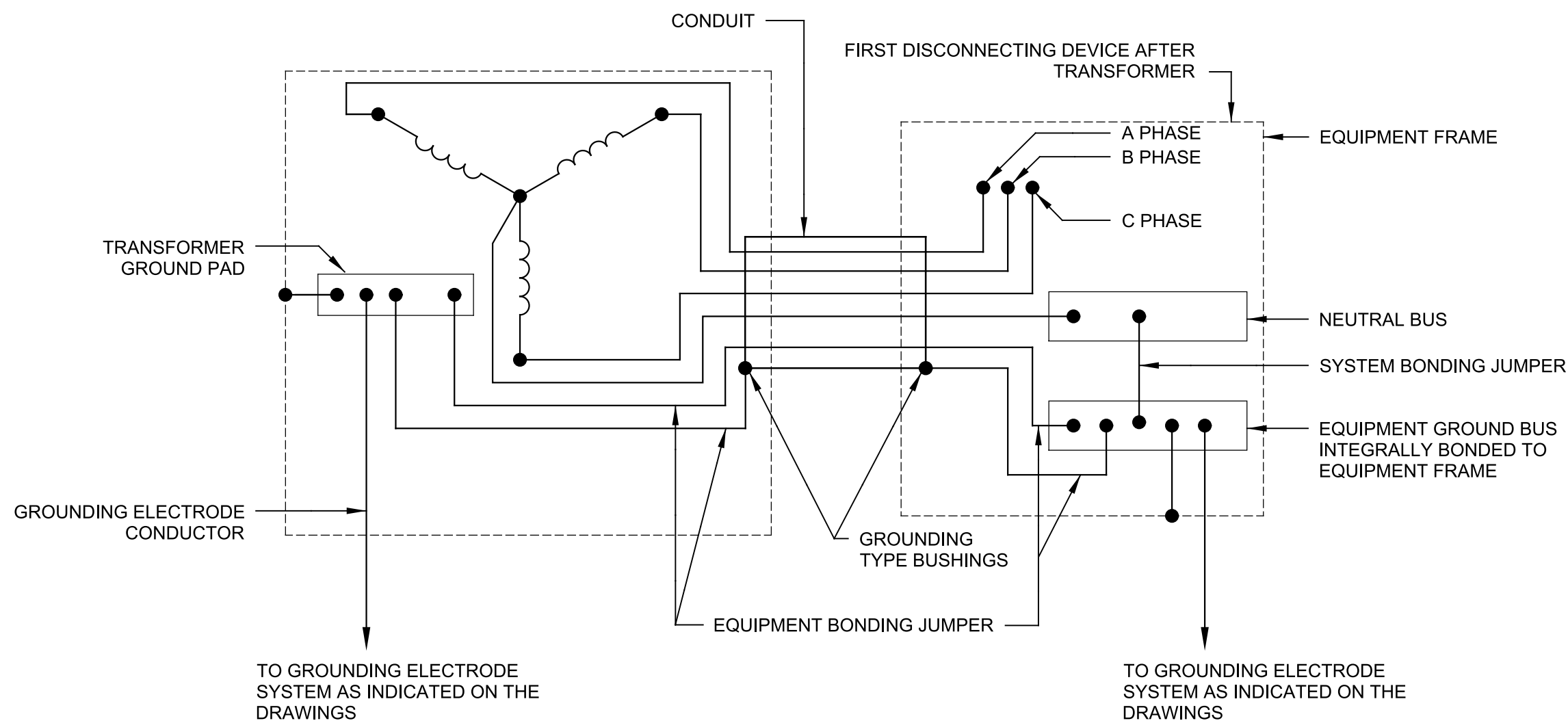
- # KEY NOTES:
- UTILITY METERING NOT SHOWN FOR CLARITY.

LEGEND:

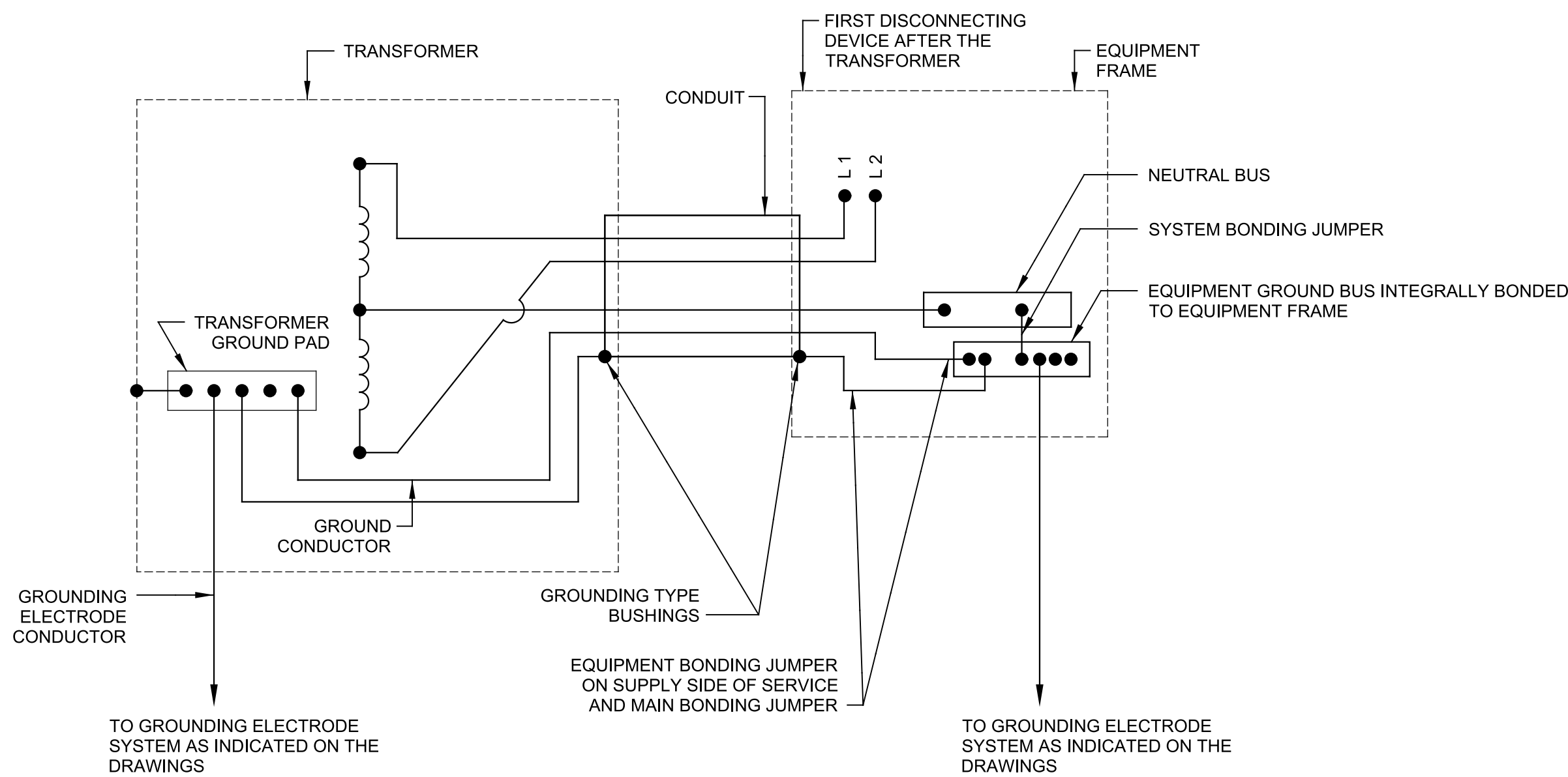
(E) - ELECTRIC OPERATOR

(S) - SHUNT TRIP

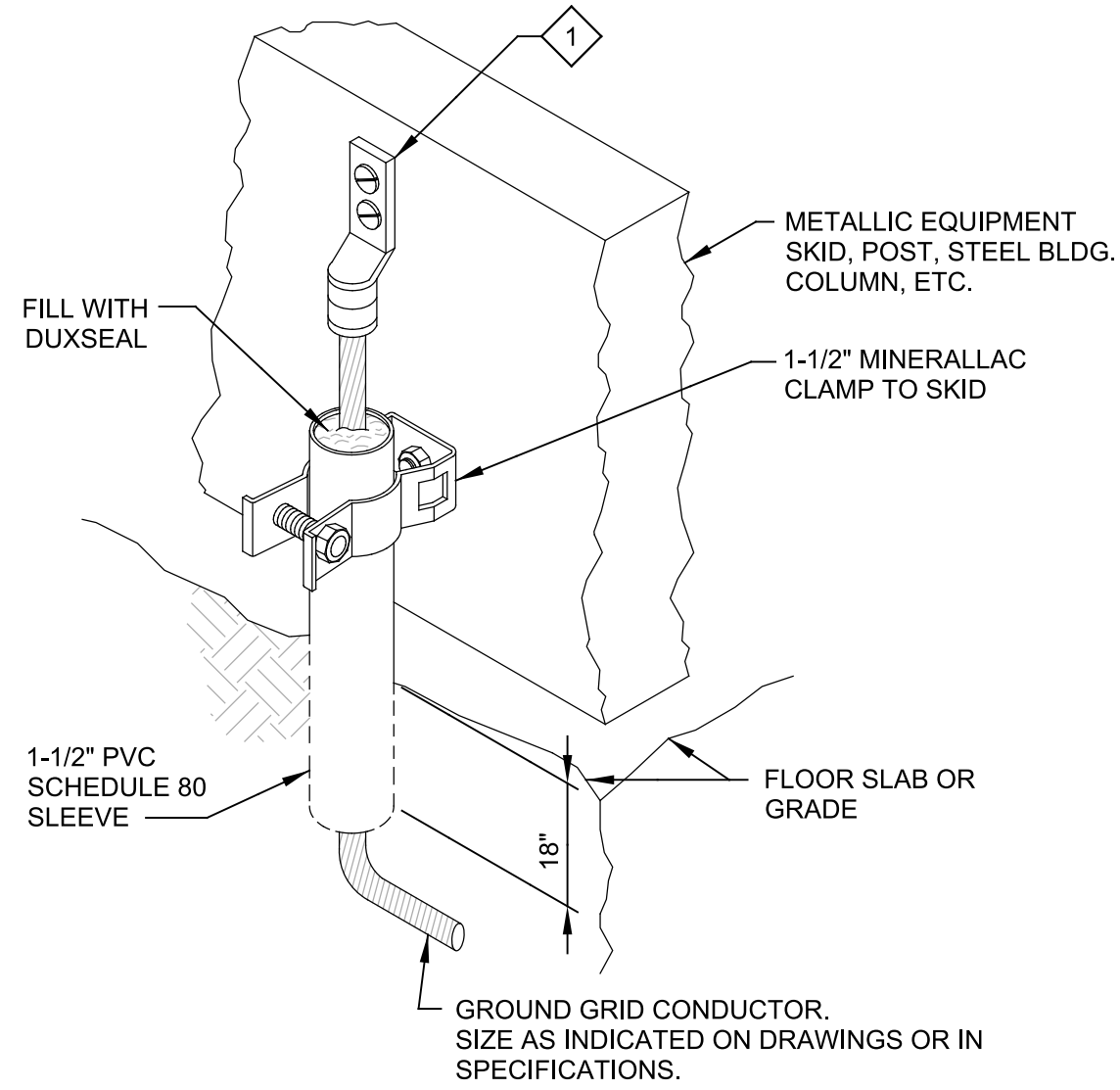
EG504 TYP S SERVICE ENTRANCE/NOT SEPARATELY DERIVED GENERATOR GROUNDING/GROUND FAULT DETAIL



EG511 TYP S TRANSFORMER (NON-UTILITY) 3-PHASE, 4-WIRE SECONDARY POWER GROUNDING DETAIL



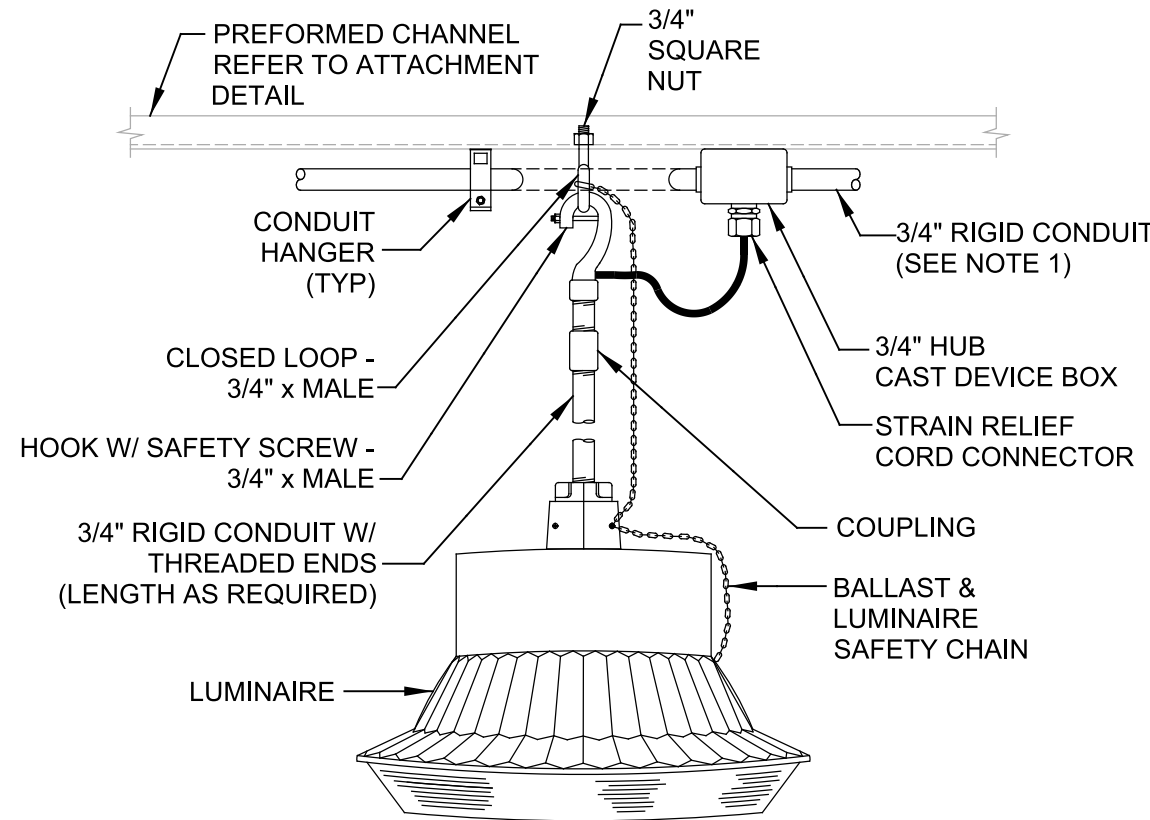
EG512 TYP S 1-PHASE TRANSFORMER (NON-UTILITY) GROUNDING DETAIL



- GENERAL NOTES:
- CLEAN ALL PAINTED SURFACES TO BARE METAL BEFORE BOLTING ON THE GROUND LUG.
 - NOTE: BOLT GROUND LUG ONLY TO EQUIPMENT SKIDS OR STRUCTURE, NOT TO EQUIPMENT OR TANK SHELLS.

- # KEY NOTES:
- HEAVY DUTY COMPRESSION TERMINAL.

EG201 TYP S EQUIPMENT SKID GROUNDING COMPRESSION CONNECTION



- GENERAL NOTES:
- AT LUMINAIRE ATTACHMENT, PROVIDE CONDUIT BEND AROUND LUMINAIRE CLOSED LOOP ASSEMBLY.

EL100 TYP S PENDANT LUMINAIRE MOUNTING

REV	DATE	KG	BY	DESCRIPTION
1	07-03-24	KG	BY	ADDENDUM NO. 6

DESIGNED KG	DRAWN DM	CHECKED TDH	DATE APR 2024
THIS WORK WAS PREPARED BY ME OR UNDER MY SUPERVISION AND CONSTRUCTION OF THIS PROJECT WILL BE UNDER MY OBSERVATION			

VERIFY SCALES

BAR IS ONE INCH ON ORIGINAL DRAWING

0 1"

IF NOT ONE INCH ON THIS SHEET, ADJUST SCALES ACCORDINGLY

carollo



COUNTY OF HAWAII

HILO WWTP REHABILITATION AND REPLACEMENT PROJECT PHASE 1

TYPICAL DETAILS

TYPICAL DETAILS - ELECTRICAL

JOB NO.
201400

DRAWING NO.
00-T-07-703

SHEET NO.
978 of 1000

1.0 COMPRESSION CONNECTORS

SPECIFICATION – 16060-2.02.A GROUNDING AND BONDING



Copper, Code, 1-Hole, Long Barrel, No Inspection Window**TYPES YA, YA-TC HYLUG™ Uninsulated Copper Compression Terminal****UL Listed 90° C, Up to 35 kV ♦**

45° and 90° angles available. Please contact Customer Service to order: 1-800-346-4175

Features & Benefits

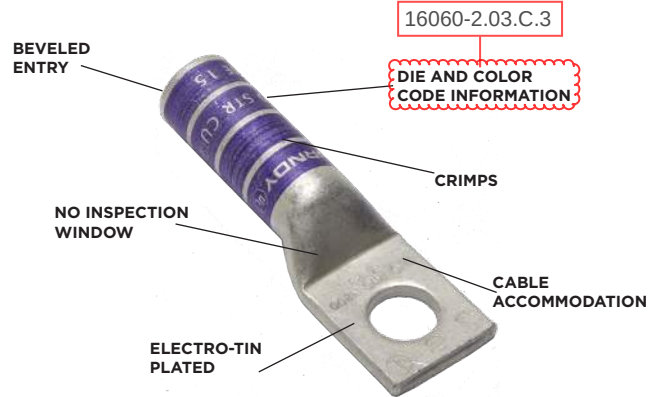
- No inspection window is ideal for more corrosive environments as the barrel transition is not open
- Barrel is designed with an internal chamfer at the wire entry to ensure smooth insertion of the wire, preventing possible damaging of the strands
- Long barrel allows for an increased number of crimps which will increase the mechanical strength of the connection
- 45° and 90° angular lugs are available; please contact Customer Service
- Electro-tin plated unless otherwise specified to reduce galvanic corrosion (bimetallic) and resist corrosive elements
- Connectors are clearly marked with stamping and barrel color coding
- BURNDY certifies its products using the BURNDY Engineered System and select other crimp tool manufacturers in accordance with one or more of the UL categories in order to obtain a UL Listing
- Terminals and splices accommodating code wire: Accommodates Code Class B (Concentric, Compressed, Compact) and Class C wire
- For applications greater than 2000 Volts consult cable manufacturer for voltage stress relief instructions

16060-2.03.C.2

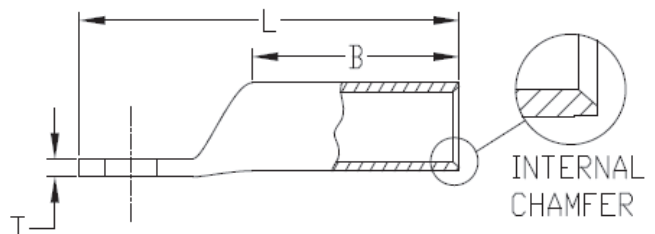
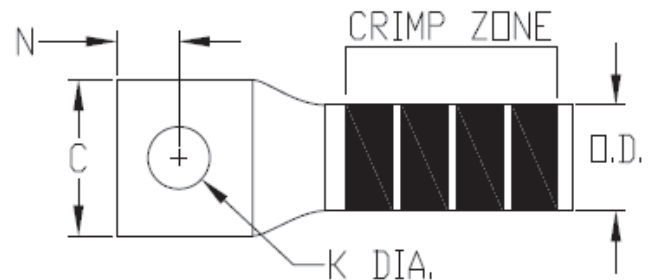
Accessories

- Hardware - Section F
- **PENETROX™ Oxide Inhibitor - Section F**
- Heat and Cold Shrink Tubing - Section D
- WIREMIKE™ Wire Micrometer - Section F
- Wire Management (Cable Ties) - Section G

16060-2.03.C.4

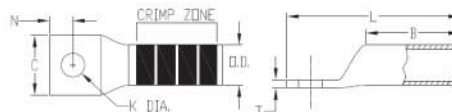


NOTE: Field bending straight lugs to achieve an angular connection will void UL Listing / CSA Certifications and BURNDY will not be liable for the connection.



Copper, Code, 1-Hole, Long Barrel, No Inspection Window

TYPES YA, YA-TC (Continued)



Notes: All dimensions shown are for reference only.

*Use PUADP1 adapter with the use of U dies in 46 Series Crimp Tools

**P-RT die sets are for use in 46 Series ONLY, no adapter required

***Y122CMR tool for use on 10 AWG - 2 AWG wire sizes ONLY

†Due to the handle force requirements to crimp a connector, this die is not recommended for manual crimp tools

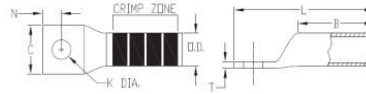
● The MM² conductor sizes listed are the recommendations for Class 2 conductor

† The MM² conductor sizes listed are the recommendations for Class 2 and Class 5 conductor

Catalog Number	Conductor			Stud Size	Figure Dimensions					Installation Tooling (# Crimps if over 1)					Wire Strip Length (IN)
	Code	Flex	Metric MM ² ●		Pad Width (C)	Barrel Length (B)	Pad Thickness (T)	Overall Length (L)	Outside Diameter (OD)	Color Code	Die Index	Dieless	MD6, MD7, 500 Series	35, 750, 46* Series	
YA8CTC10	8 AWG	#8 G,H,I,K,M DLO	8.37 † 10	#10	0.41	0.81	0.08	1.57	0.27	Red	49	Y122CMR (2) MRC840 (2) MY29 Series (2) Y8MRB1 Y1MRTC (2) 81K Series	W8CVT (2) W8CRT (2) X8CRT (2)	U8CRT (2)	7/8
YA8CTC14				1/4	0.44	0.81	0.08	1.69	0.27						
YA8CTC38				3/8	0.58	0.81	0.06	1.88	0.27						
YA6CTC8	6 AWG 6 Solid	—	—	#8	0.42	1.12	0.09	1.83	0.29	Blue	7 or 374	Y122CMR (2) MRC840 (2) MY29 Series (2) Y1MRTC (2) 81K Series	W5CRT (2) W5CRT (2) X5CRT (2) X8CART (2)	U5CRT (2) U8CABT (2)	1-3/16
YA6CTC10				#10	0.42	1.12	0.09	1.89	0.29						
YA6C				1/4	0.41	1.12	0.09	1.81	0.29						
YA6CN				1/2	0.83	1.12	0.12	2.64	0.29						
YA6CTC516				5/16	0.52	1.12	0.07	2.08	0.29						
YA6CTC38				3/8	0.58	1.12	0.06	2.21	0.29						
YA5C	5 AWG	—	—	1/4	0.44	1.12	0.07	1.98	0.30	Blue	7 or 374		W5CRT (2) W5CRT (2) X5CRT (2)	U5CRT (2)	1-3/16
YA5CN				1/2	0.83	1.12	0.12	2.67	0.30						
YA4CTC10	4 AWG	—	—	#10	0.50	1.12	0.09	1.94	0.34	Gray	8 or 346		W4CVT (2) W4CRT (2) X4CRT (2)	U4CRT (2) U6CABT (2)	1-3/16
YA4C				1/4	0.50	1.12	0.09	1.87	0.34						
YA4CTC38				3/8	0.58	1.12	0.08	2.25	0.34						
YA4CN				1/2	0.83	1.12	0.12	2.69	0.34						
YA3CTC14	3 AWG 2 Solid	—	25	1/4	0.55	1.25	0.09	2.23	0.38	White	9	Y1MRTC (4) Y122CMR (4) MY29 Series (2) 644 Series 444 Series 81K Series (2)	W3CRT (2)	U3CRT (2)	1-5/16
YA3C				5/16	0.55	1.25	0.09	2.30	0.38						
YA3CTC38				3/8	0.58	1.25	0.08	2.42	0.38						
YA3CN				1/2	0.83	1.25	0.12	2.86	0.38						
YA2CTC10	2 AWG	—	35	#10	0.60	1.25	0.11	2.10	0.42	Brown	10		W2CVT (2) W2CRT (2) X2CRT (2)	U2CRT (2)	1-5/16
YA2CTC14				1/4	0.60	1.25	0.11	2.23	0.42						
YA2C				5/16	0.60	1.25	0.11	2.29	0.42						
YA2CTC38				3/8	0.60	1.25	0.11	2.41	0.42						
YA2CN				1/2	0.83	1.25	0.12	2.88	0.42						
YA1CTC10	1 AWG	—	50	#10	0.68	1.38	0.10	2.27	0.46	Green	11	Y1MRTC (4) MRC840 (2) MY29 Series (2) 644 Series 444 Series 81K Series (2)	W1CVT (2) W1CRT (2) X1CRT (2)	U1CRT (2) U4CABT (2)	1-7/16
YA1CTC14				1/4	0.68	1.38	0.10	2.39	0.46						
YA1C				5/16	0.68	1.38	0.10	2.45	0.46						
YA1CTC38				3/8	0.68	1.38	0.10	2.58	0.46						
YA1CN				1/2	0.83	1.38	0.12	3.06	0.46						
YA25TC10	1/0 AWG	—	—	#10	0.75	1.38	0.12	2.30	0.51	Pink	12	MRC840 (4) MY29 Series (2) 644 Series 444 Series	W25VT (4)† W25RT (4)† X25RT (4)	U25RT (2) U2CABT (2)	1-7/16
YA25				5/16	0.75	1.38	0.13	2.48	0.51						
YA25TC38				3/8	0.75	1.38	0.12	2.61	0.51						
YA25N				1/2	0.83	1.38	0.11	3.05	0.51						
YA26TC14	2/0 AWG	—	70	1/4	0.83	1.50	0.12	2.58	0.56	Black	13	81K Series (2)	W26VT (4)† W26RT (4)† X26RT (4)	U26RT (2)	1-9/16
YA26TC516				5/16	0.83	1.50	0.12	2.65	0.56						

Copper, Code, 1-Hole, Long Barrel, No Inspection Window

TYPES YA, YA-TC (Continued)



Catalog Number	Conductor			Stud Size	Figure Dimensions					Installation Tooling (# Crimps if over 1)					Wire Strip Length (IN)		
	Code	Flex	Metric MM²		Pad Width (C)	Barrel Length (B)	Pad Thickness (T)	Overall Length (L)	Outside Diameter (OD)	Color Code	Die Index	Dieless	MD6, MD7, 500 Series	35, 750 46* Series			
YA26	2/0 AWG	—	70	3/8	0.83	1.50	0.12	2.77	0.56	Black	13		W26VT (4)± W26RT (4)± X26RT (4)	U26RT (2)	1-9/16		
YA26N				1/2	0.83	1.50	0.12	3.21	0.56								
YA27	3/0 AWG	—	—	1/2	0.91	1.50	0.13	3.06	0.62	Orange	14	MRC840 (4) MY29 Series (2) 644 Series 444 Series	W27VT (4)± W27RT (4)± X27RT (6)	U27RT (2)	1-9/16		
YA28TC38	4/0 AWG	—	—	3/8	1.02	1.62	0.14	2.98	0.69	Purple	15	81K Series (2)	W28VT (4)± W28RT (4)± X28RT (6)	U28RT (2)	1-11/16		
YA28				1/2	1.02	1.62	0.14	3.23	0.69								
YA28N				1/2	1.02	1.62	0.14	3.41	0.69								
YA29	250 kcmil	—	—	1/2	1.11	1.62	0.16	3.26	0.75	Yellow	16	MY29 Series (2) 644 Series 444 Series 81K Series (2)	W29VT (4)± W29RT (4)± X29RT (4)	U29RT (2)	1-11/16		
YA30	300 kcmil	—	150	1/2	1.20	2.00	0.16	3.69	0.81	White	17 or 298	644 Series 444 Series 81K Series (3)	W30VT (4)± W30RT (4)±	U30RT (4) U28ART (4)	2-1/16		
YA30N				1/2	1.20	2.00	0.16	3.88	0.81								
YA31	350 kcmil	—	185	1/2	1.29	2.00	0.18	3.73	0.88	Red	18 or 324		W31VT (4)± W31RT (4)±	U31RT (4) U29ART (4)	2-1/16		
YA32N	400 kcmil	—	—	1/2	1.40	2.12	0.19	4.09	0.95	Blue	19 or 470		W32VT (4)± W32RT (4)±	U32RT (4) U30ART (4)	2-3/16		
YA32				5/8	1.40	2.12	0.19	4.15	0.95								
YA33N	450 kcmil	—	—	1/2	1.48	2.13	0.21	4.17	1.01	Gray	326 or 538		W33VT (4)± W33RT (4)±	U33RT (4)	2-5/16		
YA33				5/8	1.48	2.13	0.21	4.24	1.01								
YA34N	500 kcmil	—	240	1/2	1.55	2.25	0.23	4.32	1.06	Brown	20 or 299		W34VT (4)± W34RT (4)±	U34RT (4) U31ART (4)	2-5/16		
YA34				5/8	1.55	2.25	0.23	4.39	1.06								
YA36N	600 kcmil	—	300	1/2	1.74	2.69	0.27	4.83	1.19	Green	22 or 472		—	U36RT (4) U32ART (4)	2-3/4		
YA36				5/8	1.74	2.69	0.27	4.90	1.19								
YA37N	650 kcmil	—	—	1/2	1.80	2.81	0.27	4.98	1.23	Orange	23			U37RT (4)	2-7/8		
YA37				5/8	1.80	2.81	0.27	5.05	1.23								
YA38N	700 kcmil	—	—	1/2	1.84	2.81	0.27	5.01	1.25	Pink	400	644 Series 444 Series 81K Series (4)		U38RT (4)	2-7/8		
YA38				5/8	1.84	2.81	0.27	5.07	1.25								
YA39N	750 kcmil	—	—	1/2	1.91	2.88	0.27	5.11	1.30	Black	24			U39RT (4) P39RT (4)	2-15/16		
YA39				5/8	1.91	2.88	0.27	5.17	1.30								
YA40	800 kcmil	—	400	5/8	1.98	2.94	0.30	5.25	1.35	Orange	25			P40RT (4)	3		
YA41N	850 kcmil	—	—	1/2	2.01	2.94	0.31	5.20	1.38	Gold	26			—	3		
YA41				5/8	2.01	2.94	0.31	5.26	1.38								
YA44N	1000 kcmil	—	500	1/2	2.19	3.00	0.33	5.38	1.50	White	27			P44RT (4)	3-1/16		
YA44				5/8	2.19	3.00	0.33	5.45	1.50								
YA45	1250 kcmil	—	—	3/4	2.46	3.19	0.38	5.93	1.69	Yellow	29			P45RT (6)	3-1/4		
YA46N	1500 kcmil	—	—	1/2	2.69	3.19	0.40	5.79	1.84	Green	31			P46RT (6)	3-5/16		
YA46				3/4	2.69	3.19	0.40	6.04	1.84								
YA47N	1750 kcmil	—	—	1/2	2.90	3.44	0.46	6.13	1.98	Gray	33			—	3-1/2		
YA47				3/4	2.90	3.44	0.46	6.38	1.98								
YA48N	2000 kcmil	—	1000	1/2	3.10	3.44	0.46	6.22	2.13	Brown	34			—	3-1/2		
YA48				3/4	3.10	3.44	0.46	6.47	2.13								

COMPRESSION CONNECTIONS

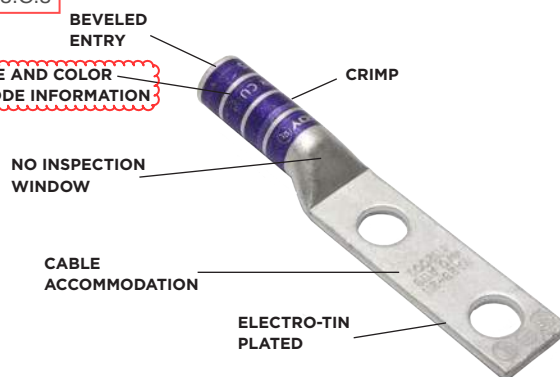
Copper, Code, 2-Hole, Long Barrel, No Inspection Window**TYPES YA-2N, YA-2TC, YA-4N HYLUG™****Uninsulated Copper Compression Terminal UL Listed 90° C, Up to 35 kV ♦**

45° and 90° angles available. Please contact Customer Service to order: 1-800-346-4175

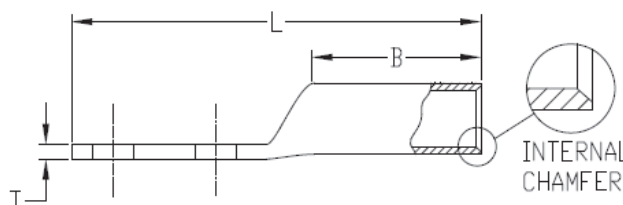
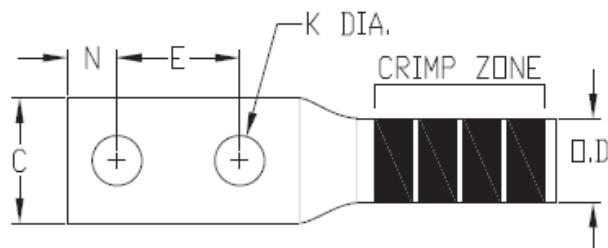
Features & Benefits

- No inspection window is ideal for more corrosive environments as the barrel transition is not open
- Barrel is designed with an internal chamfer at the wire entry to ensure smooth insertion of the wire, preventing possible damaging of the strands
- Long barrel allows for an increased number of crimps which will increase the mechanical strength of the connection
- Two hole tongue/tang is recommended when space permits as the 2-hole feature minimizes the terminations from loosening or rotating under vibration, movement, or heat cycling
- 45° and 90° angular lugs are available; please contact Customer Service
- Electro-tin plated unless otherwise specified to reduce galvanic corrosion (bimetallic) and resist corrosive elements
- Connectors are clearly marked with stamping and barrel color coding
- BURNDY certifies its products using the BURNDY Engineered System and select other crimp tool manufacturers in accordance with one or more of the UL categories in order to obtain a UL Listing
- Terminals and splices accommodating code wire: Accommodates Code Class B (Concentric, Compressed, Compact) and Class C wire
- For applications greater than 2000 Volts consult cable manufacturer for voltage stress relief instructions

16060-2.03.C.3



NOTE: Field bending straight lugs to achieve an angular connection will void UL Listing / CSA Certifications and BURNDY will not be liable for the connection.

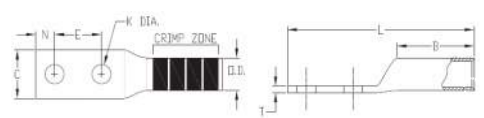
**Accessories**

- Hardware - Section F
- **PENETROX™ Oxide Inhibitor - Section F**
- Heat and Cold Shrink Tubing - Section D
- WIREMIKE™ Wire Micrometer - Section F
- Wire Management (Cable Ties) - Section G

16060-2.03.C.4

Copper, Code, 2-Hole, Long Barrel, No Inspection Window

TYPES YA-2N, YA-2TC, YA-4N (Continued)



Notes: All dimensions shown are for reference only.

*Use PUADP1 adapter with the use of U dies in 46 Series Crimp Tools

**P-RT die sets are for use in 46 Series ONLY, no adapter required

***Y122CMR tool for use on 10 AWG - 2 AWG wire sizes ONLY

‡Due to the handle force requirements to crimp a connector, this die is not recommended for manual crimp tools

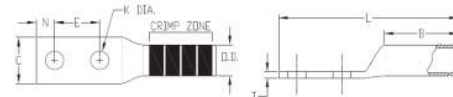
● The MM² conductor sizes listed are the recommendations for Class 2 conductor

† The MM² conductor sizes listed are the recommendations for Class 2 and Class 5 conductor

Catalog Number	Conductor			Stud Size	Figure Dimensions						Installation Tooling (# Crimps if over 1)					Wire Strip Length (IN)
	Code	Flex	Metric MM ² ●		Hole Spacing (E&F)	Pad Width (C)	Barrel Length (B)	Pad Thickness (T)	Overall Length (L)	Outside Diameter (OD)	Color Code	Die Index	Dieless	MD6, MD7, 500 Series	35, 750, 46* Series	
YA8C2TC14	8 AWG	G,H,I,K,M DLO	† 10	1/4	5/8	0.44	0.81	0.08	2.33	0.27	Red	49	Y8MRBI (2)	W8CVT (2) W8CRT (2) X8CRT (2)	U8CRT (2)	7/8
YA8C2TC14E2				1/4	3/4	0.44	0.81	0.08	2.45	0.27			YIMRTC (2)			
YA8C2TC38				3/8	1	0.58	0.81	0.06	2.89	0.27			Y122CMR (2)			
YA8C2N				1/2	1-3/4	0.83	0.81	0.12	4.08	0.27			MY29 Series (2)			
YA6C2TC14	6 AWG 6 SOL	—	—	1/4	5/8	0.45	1.12	0.08	2.65	0.29	Blue	7 or 374	MRC840 (2)	W5CVT (2) W5CRT (2) X5CRT (2)	U5CRT (2) U8CABT (2)	1-3/16
YA6C2TC14E2				1/4	3/4	0.45	1.12	0.08	2.78	0.29			Y122CMR (2)			
YA6C2TC14E1				1/4	1	0.45	1.12	0.08	3.03	0.29			YIMRTC (2)			
YA6C2TC38E2				3/8	3/4	0.58	1.12	0.06	2.97	0.29			MY29 Series (2)			
YA6C2TC38E6				3/8	7/8	0.58	1.12	0.06	3.09	0.29			MRC840 (2)			
YA6C2TC38				3/8	1	0.58	1.12	0.06	3.22	0.29			81K Series			
YA6C2N				1/2	1-3/4	0.83	1.12	0.12	4.40	0.29						
YA5C2N	5 AWG	—	—	1/2	1-3/4	0.83	1.12	0.12	4.43	0.30	Blue	7 or 374	MY29 Series (2)	W5CVT (2)	U5CRT (2)	1-3/16
YA4C2TC14	4 AWG	—	—	1/4	5/8	0.50	1.12	0.09	2.70	0.34	Gray	8 or 346	MRC840 (2)	W4CVT (2) W4CRT (2) X4CRT (2)	U4CRT (2) U6CABT (2)	1-3/16
YA4C2TC14E2				1/4	3/4	0.50	1.12	0.09	2.83	0.34			Y122CMR (4)			
YA4C2TC38				3/8	1	0.58	1.12	0.08	3.26	0.34			YIMRTC (4)			
YA4C2N				1/2	1-3/4	0.83	1.12	0.12	4.45	0.34			MY29 Series (2)			
YA3C2TC14	3 AWG 2 SOL	—	25	1/4	5/8	0.55	1.25	0.09	2.87	0.38	White	9	MRC840 (2)	W3CRT (2)	U3CRT (2)	1-5/16
YA3C2TC14E2				1/4	3/4	0.55	1.25	0.09	2.99	0.38			Y122CMR (4)			
YA3C2TC38E2				3/8	3/4	0.58	1.25	0.08	3.18	0.38			YIMRTC (4)			
YA3C2TC38				3/8	1	0.58	1.25	0.08	3.43	0.38			MY29 Series (2)			
YA3C2N				1/2	1-3/4	0.83	1.25	0.12	4.62	0.38			MRC840 (2)			
YA2C2TC14	2 AWG	—	35	1/4	5/8	0.60	1.25	0.11	2.86	0.42	Brown	10	644 Series	W2CVT (2) W2CRT (2) X2CRT (2)	U2CRT (2)	1-5/16
YA2C2TC14E2				1/4	3/4	0.60	1.25	0.11	2.98	0.42			444 Series			
YA2C2TC516E2				5/16	3/4	0.60	1.25	0.11	3.05	0.42			81K Series (2)			
YA2C2TC38E2				3/8	3/4	0.60	1.25	0.11	3.17	0.42			Y122CMR*** (4)			
YA2C2TC38E6				3/8	7/8	0.60	1.25	0.11	3.30	0.42			YIMRTC (4)			
YA2C2TC38				3/8	1	0.60	1.25	0.11	3.42	0.42			MY29 Series (2)			
YA2C2NTC38				3/8	1-3/4	0.60	1.25	0.11	4.17	0.42			MRC840 (2)			
YA2C2N				1/2	1-3/4	0.83	1.25	0.12	4.64	0.42			644 Series			
YA1C2TC14	1 AWG	—	50	1/4	5/8	0.67	1.38	0.10	3.03	0.46	Green	11 or 375	81K Series (2)	W1CVT (2) W1CRT (2) X1CRT (2)	U1CRT (2) U4CABT (2)	1-7/16
YA1C2TC14E2				1/4	3/4	0.67	1.38	0.10	3.15	0.46						
YA1C2TC38				3/8	1	0.67	1.38	0.10	3.59	0.46						
YA1C2N				1/2	1-3/4	0.83	1.38	0.12	4.82	0.46						
YA252TC14	1/0 AWG	—	—	1/4	5/8	0.75	1.38	0.12	3.05	0.51	Pink	12	MY29 Series (2)	W25VT (4)‡ W25RT (4)‡ X25RT (4)	U25RT (2) U2CABT (2)	1-7/16
YA252TC14E2				1/4	3/4	0.75	1.38	0.12	3.18	0.51			MRC840 (4)			
YA252TC516				5/16	1	0.75	1.38	0.12	3.49	0.51			644 Series			
YA252TC38				3/8	1	0.75	1.38	0.12	3.62	0.51			444 Series			
YA252NTC38				3/8	1-3/4	0.75	1.38	0.12	4.37	0.51			81K Series (2)			
YA252N				1/2	1-3/4	0.83	1.38	0.11	4.81	0.51						

Copper, Code, 2-Hole, Long Barrel, No Inspection Window

TYPES YA-2N, YA-2TC, YA-4N (Continued)



Notes: All dimensions shown are for reference only.

*Use PUADPI adapter with the use of U dies in 46 Series Crimp Tools

**P-RT die sets are for use in 46 Series ONLY, no adapter required

***Y122CMR tool for use on 10 AWG - 2 AWG wire sizes ONLY

†Due to the handle force requirements to crimp a connector, this die is not recommended for manual crimp tools

● The MM² conductor sizes listed are the recommendations for Class 2 conductor

† The MM² conductor sizes listed are the recommendations for Class 2 and Class 5 conductor

Catalog Number	Conductor			Stud Size	Figure Dimensions						Installation Tooling (# Crimps if over 1)					Wire Strip Length (IN)
	Code	Flex	Metric MM ² ●		Hole Spacing (E&F)	Pad Width (C)	Barrel Length (B)	Pad Thickness (T)	Overall Length (L)	Outside Diameter (OD)	Color Code	Die Index	Dieless	MD6, MD7, 500 Series	35, 750, 46* Series	
YA262TC14	2/0 AWG	—	70	1/4	5/8	0.83	1.50	0.12	3.22	0.56	Black	13	MY29 Series (2) MRC840 (4) 644 Series 444 Series 81K Series (2)	W26VT (4)± W26RT (4)± X26RT (4)	U26RT (2)	1-9/16
YA262TC14E2				1/4	3/4	0.83	1.50	0.12	3.34	0.56						
YA262TC38				3/8	1	0.83	1.50	0.12	3.78	0.56						
YA262N				1/2	1-3/4	0.83	1.50	0.12	4.97	0.56						
YA272TC14E2	3/0 AWG	—	—	1/4	3/4	0.90	1.50	0.12	3.38	0.62	Orange	14	MY29 Series (2) MRC840 (4) 644 Series	W27VT (4)± W27RT (4)± X27RT (6)	U27RT (2)	1-9/16
YA272TC38				3/8	1	0.90	1.50	0.12	3.82	0.62						
YA272N				1/2	1-3/4	0.91	1.50	0.13	5.01	0.62						
YA282TC14E2				1/4	3/4	1.02	1.62	0.14	3.55	0.69						
YA282TC38	4/0 AWG	—	—	3/8	1	1.02	1.62	0.14	3.99	0.69	Purple	15	444 Series 81K Series (2)	W28VT (4)± W28RT (4)± X28RT (6)	U28RT (2)	1-11/16
YA282NTC38				3/8	1-3/4	1.02	1.62	0.14	4.74	0.69						
YA282N				1/2	1-3/4	1.02	1.62	0.14	5.17	0.69						
YA292TC38	250 kcmil	—	—	3/8	1	1.10	1.62	0.16	4.02	0.75	Yellow	16	MY29 Series (2) 644 Series 444 Series 81K Series (2)	W29VT (4)± W29RT (4)± X29RT (8)	U29RT (2)	1-11/16
YA292N				1/2	1-3/4	1.11	1.62	0.16	5.21	0.75						
YA292TC58E16				5/8	1-3/4	1.11	1.62	0.16	5.27	0.75						
YA302TC38	300 kcmil	—	150	3/8	1	1.20	2.00	0.16	4.45	0.81	White	17 or 298	644 Series 444 Series 81K Series (3)	W30VT (4)± W30RT (4)±	U30RT (4) U28ART (4)	2-1/16
YA302N				1/2	1-3/4	1.20	2.00	0.16	5.64	0.81						
YA312TC14E2	350 kcmil	—	185	1/4	3/4	1.29	2.00	0.18	4.05	0.88	Red	18 or 324		W31RT (4)± W31VT (4)±	U31RT (4) U29ART (4)	2-1/16
YA312TC38				3/8	1	1.29	2.00	0.18	4.49	0.88						
YA312N				1/2	1-3/4	1.29	2.00	0.18	5.69	0.88						
YA322TC38	400 kcmil	—	—	3/8	1	1.40	2.12	0.19	4.66	0.95	Blue	19 or 470		W32VT (4)± W32RT (4)±	U32RT (4) U30ART (4)	2-3/16
YA322N				1/2	1-3/4	1.40	2.12	0.19	5.85	0.95						
YA332N	450 kcmil	—	—	1/2	1-3/4	1.48	2.13	0.21	5.93	1.01	Gray	326 or 538		W33VT (4)± W33RT (4)±	U33RT (4)	2-3/16
YA342TC14E2	500 kcmil	—	240	1/4	3/4	1.55	2.25	0.23	4.46	1.06	Brown	20 or 299		W34VT (4)± W34RT (4)±	U34RT (4)	2-5/16
YA342TC38				3/8	1	1.55	2.25	0.23	4.90	1.06						
YA342N				1/2	1-3/4	1.55	2.25	0.23	6.06	1.06						
YA352N	550 kcmil	—	—	1/2	1-3/4	1.65	2.63	0.25	6.49	1.13	Yellow	21	644 Series 444 Series 81K Series (4)	—	U35RT (4)	2-11/16
YA362TC38	600 kcmil	—	300	3/8	1	1.74	2.69	0.26	5.59	1.19	Green	22 or 472		—	U36RT (4) U32ART (4)	2-3/4
YA362N				1/2	1-3/4	1.74	2.69	0.27	6.59	1.19						
YA372N	650 kcmil	—	—	1/2	1-3/4	1.80	2.81	0.27	6.74	1.23	Orange	23		—	U37RT (4)	2-7/8
YA382TC38	700 kcmil	—	—	3/8	1	1.84	2.81	0.27	5.76	1.25	Pink	400		—	U38RT (4)	2-7/8
YA382N				1/2	1-3/4	1.84	2.81	0.27	6.77	1.25						

TYPE YGL-C

HYGRID™ CROSS CONNECTOR

An irreversible compression ground grid cross connector which allows adjustment of the compression elements prior to installation. Only six connectors and four dies are required to install all combinations from #6 solid through 500 kcmil. UL467 Listed. Acceptable for direct burial in earth and concrete. Prefilled with PENETROX™ compound and strip sealed.

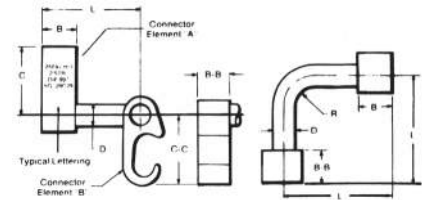


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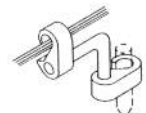


IEEE-837*

16060-2.03.C.4



Ground Grid Cross Connection



Cable to Ground Rod

✓ UL96 Listed for Lightning Protection.



Catalog Number	B	BB	C	C-C	D	L	R
YGL2C2	0.75	0.75	1.09	1.09	0.31	2.50	0.31
YGL29C2			1.66	1.09	0.31		0.31
YGL29C29			1.66	1.66	0.50		0.50
YGL34C2			2.09	1.09	0.31		0.31
YGL34C29			2.09	1.66	0.50		0.50
YGL34C34	1.10	1.10	2.28	2.28	0.75		0.75

Catalog Number	Cable to Cable		Cable to Ground Rod		To Rebar
	Element "A"	Element "B"	Element "A"	Element "B"	Element "A"
YGL2C2	#6 Sol. (0.162) - #2 Str. (0.292) {59500} - {59500}	#6 Sol. (0.162) - #2 Str. (0.292) {59500} - {59500}	—	—	—
YGL29C2	#1 Str. (0.332) - 250 kcmil (0.575) {98500} - {131500}	#6 Sol. (0.162) - #2 Str. (0.292) {59500} - {59500}	1/2" - 5/8" Rod	#6 Sol. (0.162) - #2 Str. (0.292)	3/8" - 1/2"
YGL29C29	#2 Str. (0.292) - 250 kcmil (0.575) {65500} - {131500}	#2 Str. (0.292) - 250 kcmil (0.575) {65500} - {131500}	1/2" - 5/8" Rod	#2 Str. (0.292) - 250 kcmil (0.575)	#3 - 4 Rebar
YGL34C2		#6 Sol. (0.162) - #2 Str. (0.292)	5/8" - 3/4" Rod	#6 Sol. (0.162) - #2 Str. (0.292)	5/8" - 3/4" #5 - 6 Rebar
YGL34C29	250 kcmil (0.575) - 500 kcmil (0.813)	#2 Str. (0.292) - 250 kcmil (0.575)		#2 Str. (0.292) - 250 kcmil (0.575)	
YGL34C34		250 kcmil (0.575) - 500 kcmil (0.813)		250 kcmil (0.575) - 500 kcmil (0.813)	

Dimensions in brackets { } represent lightning protection conductors.

Catalog Number	Installation Tools, Die Set Catalog Number (Number of Crimps)							
	Y750/Y35/Y39 HYPRESS™		PAT750-18V		Y45 HYPRESS™		Y46 HYPRESS™	
	Element "A"	Element "B"	Element "A"	Element "B"	Element "A"	Element "B"	Element "A"	Element "B"
YGL2C2	U-O (1)	U-O (1)	U-O (1)	U-O (1)	U-O (1)	U-O (1)	U-O (1)	U-O (1)
YGL29C2	U997 (1)	U-O (1)	③ U997P (1)	U-O (1)	U997 (1)	U-O (1)	U997 (1)	U-O (1)
YGL29C29	U997 (1)	U997 (1)	③ U997P (1)	③ U997P (1)	U997 (1)	U997 (1)	U997 (1)	U997 (1)
YGL34C2 †	PU998 (1)	PU998 (1)	PU998 (1)	U-O (1)	S998 or PU998 (1)	U-O (1)	P998 or PU998 (1)	U-O (1)
YGL34C29 †	PU998 (1)	PU998 (1)	PU998 (1)	③ U997P (1)	S998 or PU998 (1)	U997 (1)	P998 or PU998 (1)	U997 (1)
YGL34C34 †	U1011 (3)	U1011 (3)	U1011 (3)	U1011 (3)	S1011 (3)	S1011 (3)	P1011 (3)	P1011 (3)

- Where a "U" or "PU" die is recommended with Y45 HYPRESS™, a PT6515 adapter must be used.
- Where a "U" or "PU" die is recommended with the Y46 HYPRESS™, a PUADP-1 adapter must be used.
- ③ Polarized die for the PAT750-18V.

† These connectors can only be installed using the Y750, Y45, or Y46 HYPRESS™ with the recommended dies.

*Please contact Customer Service for applications requiring IEEE-837 2014.

NOTES:

- Before crimping, both connector elements can be turned on rod diameter "D" to any desired position.
- Clean rust and/or protective coatings from rebar prior to installation.
- When attaching connector to ground rod, ground rod must be embossed with appropriate PRECRIMP™ die. For connections that must meet IEEE 837 requirements UPRECRIMP-type PRE crimp dies must be used for maximum clamping retention.

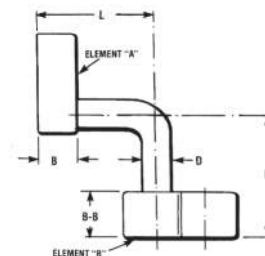
Ground Rod Dia.	PRECRIMP Dies	
1/2"	UPRECRIMP 12	U2CABT
5/8"	UPRECRIMP 58	
3/4"	UPRECRIMP 34	

TYPE YGLR-C

GRIDLOK™

High Strength Irreversible
Compression Ground Rod to Grid
Connector

16060-2.03.C.2



E-12

Ground grid connector for a wide range of copper cable to ground rod. Provides high torque strength on ground rod. UL467 Listed. Acceptable for direct burial in earth and concrete. Prefilled with PENETROX™ compound and strip sealed.

16060-2.03.C.4



IEEE-837*



Catalog Number	B	B-B	D	L	Commercial Copper Cable Range Element "A"	Metric Copper Cable Range Element "A"	CopperWeld Cable Range Element "A"	Ground Rod Dia Element "B"
YGLR29C12	0.75	0.88 [22.4]	0.31	2.53	#2 Str. (0.292 Dia.) - 250 kcmil (0.575 Dia.)	35mm ² (7.62mm Dia.) - 120 mm ² (14.40mm Dia.)	91.65 kcmil (0.343 Dia.) - 248.8 kcmil (0.572 Dia.)	1/2" [12.7]
YGLR34C12	0.75	0.88 [22.4]	0.31	2.53	250 kcmil (0.575 Dia.) - 500 kcmil (0.813 Dia.)	120 mm ² (14.40mm Dia.) - 240 mm ² (20.35mm Dia.)	248.8 kcmil (0.572 Dia.) - 498.8 kcmil (0.810 Dia.)	
YGLR29C58	0.75	0.88 [22.4]	0.31	2.53	#2 Str. (0.292 Dia.) - 250 kcmil (0.575 Dia.)	35mm ² (7.62mm Dia.) - 120 mm ² (14.40mm Dia.)	91.65 kcmil (0.343 Dia.) - 248.8 kcmil (0.572 Dia.)	5/8" [15.9]
YGLR34C58	0.75	0.88 [22.4]	0.31	2.53	250 kcmil (0.575 Dia.) - 500 kcmil (0.813 Dia.)	120 mm ² (14.40mm Dia.) - 240 mm ² (20.35mm Dia.)	248.8 kcmil (0.572 Dia.) - 498.8 kcmil (0.810 Dia.)	
YGLR29C34	0.75	0.88 [22.4]	0.50	2.63	#2 Str. (0.292 Dia.) - 250 kcmil (0.575 Dia.)	35mm ² (7.62mm Dia.) - 120 mm ² (14.40mm Dia.)	91.65 kcmil (0.343 Dia.) - 248.8 kcmil (0.572 Dia.)	3/4" [19.1]
YGLR34C34	0.75	0.88 [22.4]	0.50	2.63	250 kcmil (0.575 Dia.) - 500 kcmil (0.813 Dia.)	120 mm ² (14.40mm Dia.) - 240 mm ² (20.35mm Dia.)	248.8 kcmil (0.572 Dia.) - 498.8 kcmil (0.810 Dia.)	
YGLR29C100	0.75	0.88 [22.4]	0.50	2.63	#2 Str. (0.292 Dia.) - 250 kcmil (0.575 Dia.)	35mm ² (7.62mm Dia.) - 120 mm ² (14.40mm Dia.)	91.65 kcmil (0.343 Dia.) - 248.8 kcmil (0.572 Dia.)	1" [25.4]
YGLR34C100	0.75	0.88 [22.4]	0.50	2.63	250 kcmil (0.575 Dia.) - 500 kcmil (0.813 Dia.)	120 mm ² (14.40mm Dia.) - 240 mm ² (20.35mm Dia.)	248.8 kcmil (0.572 Dia.) - 498.8 kcmil (0.810 Dia.)	

Catalog Number	Installation Tools, Die Set Cat. No. (Number of Crimps)							
	Y35/Y39 HYPRESS™		Y750/PAT750		② Y45 HYPRESS™		③ Y46 HYPRESS™	
	Element "A"	Element "B"	Element "A"	Element "B"	Element "A"	Element "B"	Element "A"	Element "B"
YGLR29C12	U997 (1)	PU998 (1)	U997 (1)	U1011 (2) or PU998 (1)	U997 (1)	S1012 (2) or PU998 (1)	U997 (1)	P1011 (2) or PU998 (1)
YGLR34C12	—	—	U1011 or PU998 (1)	U1011 (2) or PU998 (1)	S998 or PU998 (1)	S1012 (2) or PU998 (1)	P998 or PU998 (1)	P1011 (2) or PU998 (1)
YGLR29C58	U997 (1)	PU998 (1)	U997 (1)	U1011 (2) or PU998 (1)	U997 (1)	S1012 (2) or PU998 (1)	U997 (1)	P1011 (2) or PU998 (1)
YGLR34C58	—	—	U1011 or PU998 (1)	U1011 (2) or PU998 (1)	S998 or PU998 (1)	S1012 (2) or PU998 (1)	P998 or PU998 (1)	P1011 (2) or PU998 (1)
YGLR29C34	U997 (1)	PU998 (1)	U997 (1)	U1011 (2) or PU998 (1)	U997 (1)	S1012 (2) or PU998 (1)	U997 (1)	P1011 (2) or PU998 (1)
YGLR34C34	—	—	U1011 (2) or PU998 (1)	U1011 (2) or PU998 (1)	U1011 (2) or PU998 (1)	S1012 (2) or PU998 (1)	U1011 (2) or PU998 (1)	P1011 (2) or PU998 (1)
YGLR29C100	—	—	U997 (1)	U1011 (2) or PU998 (1)	U997 (1)	S1012 (2) or PU998 (1)	U997 (1)	P1011 (2) or PU998 (1)
YGLR34C100	—	—	U1011 (2) or PU998 (1)	U1011 (2) or PU998 (1)	U1011 (2) or PU998 (1)	S1012 (2) or PU998 (1)	U1011 (2) or PU998 (1)	P1011 (2) or PU998 (1)

NOTES:

- ② Where a "U" or "PU" die is recommended with Y45 HYPRESS™, a PT6515 adapter is required
③ Where a "U" or "PU" die is recommended with Y46 HYPRESS™, a PAADP-1 adapter is required

*Please contact Customer Service for applications requiring IEEE-837 2014.

- Before crimping, both connector elements can be turned on rod diameter "D" to any desired position.
- When attaching connector to ground rod, ground rod must be embossed with appropriate PRECRIMP die. For connectors that must meet IEEE 837 requirements, UPRECRIMP-type precrimp dies must be used for maximum clamping retention. No precrimp needed if using S1012, P1011 or U1011 die sets.

Ground Rod Dia.	PRECRIMP Dies	
1/2"	UPRECRIMP 12	U2CABT
5/8"	UPRECRIMP 58	
3/4"	UPRECRIMP 34	

TYPE YGHP-C

HYTAP™

Irreversible compression ground tap figure 6 can be used as a tap connector or as a tap splice connector. Four die sets and eight connectors can accommodate a conductor range from #8 solid through 500 kcmil plus 1/2", 5/8" and 3/4" copper bonded ground rods. UL467 Listed. Acceptable for direct burial in earth and concrete. Prefilled with PENETROX™ compound and strip sealed.



✓ UL96 Listed for Lightning Protection.

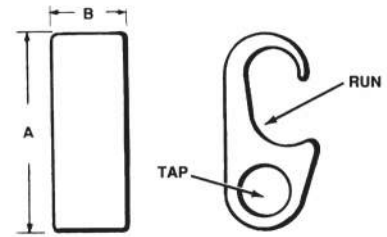


Fig. 1



Fig. 2

16060-2.03.C.4
IEEE-837*



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E-13

Catalog Number	Fig. #	Accommodates		⑨ Cable to Rebar		B	Die Index	Installation Data				
		① Run	① Tap	Run	Tap			Y750/Y35/Y39	PAT750-18V	Y45 ②	Y46 ③	# of Crimps
YGHP2C2	1	#6 Sol. (0.162) {59500} - #2 Str. (0.292) {59500}	⑥ #6 Sol. (0.162) {59500} - #2 Str. (0.292) {59500}	—	—	0.75	O	UO	UO	UO	UO	1
YGHP2C6W6W ⑦	2	#6 Sol. (0.162) - #2 Str. (0.292)	#8 Sol. (0.128) - #6 Str. (0.184) Qty. 2	—	—							1
YGHP29C6W6W ⑦	2		#8 Sol. (0.128) - #6 Str. (0.184) Qty. 2	#3 Rebar 3/8 - 1/2 #4 Rebar	#8 Sol. - 6 Str.							1
YGHP29C2	1	1/0 Str. (0.372) {98500} - 250 kcmil (0.575) {131500}	#4 Sol. (0.204) {#4 Sol.} - #2 Str. (0.292) {#2 Str.}	#3 Rebar 3/8 - 1/2 #4 Rebar	#2 Str.	2.31	997	U997	U997P ④	U997	U997	1
YGHP29C26	1	1/2" - 5/8" Rod	1/0 Str. (0.372) {98500} - 2/0 Str. (0.419) {98500}	#3 Rebar 3/8 - 1/2 #4 Rebar	1/0 Str. - 2/0 Str.							1
YGHP29C29 ⑧	1		3/0 Str. (0.470) {131500} - 250 kcmil (0.575) {211500}	#3 Rebar 3/8 - 1/2 #4 Rebar	3/0 Str. - 250 kcmil							1
YGHP34C2 ⑤	1	250 kcmil (0.575) {250 kcmil} - 500 kcmil (0.813) {500 kcmil} 5/8" - 3/4" Rod	#4 Sol. (0.204) - #2 Str. (0.292)	#5 Rebar 5/8 - 3/4 #6 Rebar	—	2.75	998	PU998	PU998	PU998 or S998	PU998 or P998	1
YGHP34C26 ⑤	1		1/0 Str. (0.372) {98500} - 2/0 Str. (0.419) {98500}	#5 Rebar 5/8 - 3/4 #6 Rebar	1/0 Str. - 2/0 Str.							1
YGHP34C29 ⑤	1		3/0 Str. (0.470) {131500} - 250 kcmil (0.575) {211500}	#5 Rebar 5/8 - 3/4 #6 Rebar	3/0 Str. - 250 kcmil							1
YGHP34C34 ⑤	1	250 kcmil (0.575) - 500 kcmil (0.813) 5/8" - 3/4" Rod	350 kcmil (0.681) - 500 kcmil (0.843)	#5 Rebar 5/8 - 3/4 #6 Rebar	350 kcmil - 500 kcmil	2.75	1011	U1011	U1011	S1011	P1011	3

NOTES:

- ① Dimensions in brackets { } represent lightning protection conductors.
- ② Where a "U" or "PU" die is recommended with Y45 HYPRESS™, a PT6515 adapter is required.
- ③ Where a "U" or "PU" die is recommended with Y46 HYPRESS™, a PUADP-1 adapter is required.
- ④ Polarized die for the PAT750 series tool.
- ⑤ These connectors can only be installed using the Y750, Y45 or Y46 series of HYPRESS™ tools; cannot be installed with the Y35 or Y39 series.

- ⑥ When using #6 Sol in tap, fold conductor double for improved fill.
- ⑦ Not UL96/CSA
- ⑧ When using 3/0 in tap, minimum run conductor must be 2/0 Str.
- ⑨ Clean rust and protective coatings from rebar prior to installation to provide proper ground connection. Precrimping is not required.

*Please contact Customer Service for applications requiring IEEE-837 2014.

Ground Rod Dia.	PRECRIMP Dies	
1/2"	UPRECRIMP 12	U2CABT
5/8"	UPRECRIMP 58	
3/4"	UPRECRIMP 34	

For increased rotational resistance on ground rods, pre-crimp ground rod with U2CABT, UPRECRIMP dies may be used for even greater rotation and vibration resistance on ground rods.

TYPE YGHC-C

HYTAP™ CONNECTOR

Irreversible compression ground tap figure "C" connectors. Accommodates all cable combinations from #6 solid through 500 kcmil. "C"- shaped opening permits placing two continuous parallel cables into conductor groove. UL 467 Listed.

Acceptable for direct burial in earth or concrete.

Prefilled with PENETROX™ compound and strip sealed. Certain sizes are also UL467 Listed and CSA Certified for wire to ground rod.

✓ UL96 Listed for Lightning Protection.

16060-2.03.C.2



16060-2.03.C.4

IEEE-837*



E-15

Catalog Number	① Commercial Copper Cable to Cable or ② Cable to Ground Rod		Stranded Copper Cable Range Metric		A	B	Die Index No.	Installation Data			No. of Crimps
	Run	Tap	Run	Tap				Y750/Y35/ PAT750/Y39	Y45 ③	Y46 ④	
YGHC2C2	#6 Sol. (0.162) #2 Str. (0.292) 1/4" Rod⑤	#6 Sol. (0.162) #2 Str. (0.292)	10 mm² (4.12 mm) 35 mm² (7.62 mm)	10 mm² (4.12 mm) 35 mm² (7.62 mm)	1.16	0.75	C	U-C	U-C	U-C	1
YGHC26C2	1 Str. (0.328) {98500} 2/0 Str. (0.419) {98500} 3/8" Rod ⑤	#6 Sol. (0.162) {#6 Sol.} #2 Str. (0.292) {#2 Str.}	35 mm² (7.62 mm) 70 mm² (10.9 mm)	10 mm² (4.12 mm) 35 mm² (7.62 mm)	1.41	0.75	O	U-O	U-O	U-O	1
YGHC26C26	1 Str. (0.328) {98500} 2/0 Str. (0.419) {98500} 3/8" Rod ⑤	1 Str. (0.328) {98500} 2/0 Str. (0.419) {98500}	35 mm² (7.62 mm) 70 mm² (10.9 mm)	35 mm² (7.62 mm) 70 mm² (10.9 mm)	1.54	0.75	O	U-O	U-O	U-O	1
YGHC29C26	3/0 Str. (0.470) {3/0 Str.} 250 kcmil (0.575) {250 kcmil} 1/2" or 5/8" Rod⑤	6 Sol. (0.162) {59500} 2/0 Str. (0.419) {98500}	95 mm² (12.5 mm) 120 mm² (14.4 mm)	10 mm² (4.10 mm) 70 mm² (10.90 mm)	1.97	0.75	997	U997	U997	U997	1
YGHC29C29	3/0 Str. (0.470) 250 kcmil (0.575) 1/2" or 5/8" Rod ⑤	3/0 Str. (0.470) 250 kcmil (0.575)	95 mm² (12.5 mm) 120 mm² (14.4 mm)	95 mm² (12.5 mm) 120 mm² (14.4 mm)	2.06	0.88	997	U997	U997	U997	1
YGHC34C26 ⑥	300 kcmil (0.630) {300 kcmil} 500 kcmil (0.813) {500 kcmil} 3/4" Rod ⑤	6 Sol. (0.162) {59500} 2/0 Str. (0.419) {98500}	150 mm² (16 mm) 240 mm² (20.35 mm)	10 mm² (4.10 mm) 70 mm² (10.90 mm)	2.42	0.88	1011	U1011	S1011	P1011	2
YGHC34C29 ⑥	300 kcmil (0.630) 500 kcmil (0.813)	3/0 Str. (0.470) 250 kcmil (0.575)	150 mm² (16 mm) 240 mm² (20.35 mm)	95 mm² (12.5 mm) 120 mm² (14.4 mm)	2.67	0.88	1011	U1011	S1011	P1011	2
YGHC34C34 ⑥	300 kcmil (0.630) 500 kcmil (0.813)	300 kcmil (0.630) 500 kcmil (0.813)	150 mm² (16 mm) 240 mm² (20.35 mm)	150 mm² (16 mm) 240 mm² (20.35 mm)	2.91	1.10	1011	U1011	S1011	P1011	3

① Dimensions in brackets { } represent lightning protection conductors

② For ground rod to wire applications, ground rod must be precrimped; see table for appropriate precrimp dies

③ Where a "U" or "PU" die is recommended with the Y45 HYPRESS™, a PT6515 adapter must be used

④ Where a "U" or "PU" die is recommended with the Y46 HYPRESS™, a PUADP-1 adapter must be used

⑤ Ground rod to copper cable is UL467 Listed for direct burial in earth and concrete

⑥ These connectors cannot be installed with the Y35 or Y39 HYPRESS™

NOTES:

Listed under UL486A for copper wire connectors

For connectors without PENETROX™ oxide inhibitor, add suffix "NP" to the end of the catalog number (example: YGHC2C2-NP)

*Please contact Customer Service for applications requiring IEEE-837 2014.

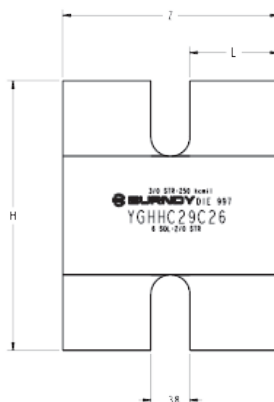
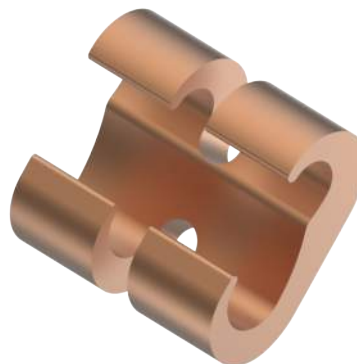
Ground Rod Dia.	PRECRIMP Dies	
1/2"	UPRECRIMP 12	U2CABT
5/8"	UPRECRIMP 58	
3/4"	UPRECRIMP 34	

TYPE YGHH-C

HYTAP™ CONNECTOR

Irreversible compression ground tap figure "C" connectors. Accommodates all cable combinations from #6 solid through 500 kcmil. "C"-shaped opening permits placing two continuous parallel cables into conductor groove. The YGHH-C series is qualified to IEEE-837 2014. Prefilled with PENETROX™ compound and strip sealed.

**IEEE-837
2014
Qualified!**



E-16

16060-2.03.C.4



IEEE-837



Catalog Number	Commercial Copper Cable to Cable or Cable to Ground Rod		Stranded Copper Cable Range Metric		H	L	Z	Die Index No.	Installation Data			No. of Crimps
	Run	Tap	Run	Tap					Y750/Y35/ PAT750/Y39	Y45 ③	Y46 ④	
YGHH-C26C26	1 Str. (0.328) 2/0 Str. (0.419) 3/8" Rod	1 Str. (0.328) 2/0 Str. (0.419)	35 mm² (7.62 mm) 70 mm² (10.9 mm)	35 mm² (7.62 mm) 70 mm² (10.9 mm)	1.54 [39]	0.75 [19]	1.88 [48]	O	U-O	U-O	U-O	2
YGHH-C29C26	3/0 Str. (0.470) 250 kcmil (0.575) 1/2" or 5/8" Rod	6 Sol. (0.162) 2/0 Str. (0.419)	95 mm² (12.5 mm) 120 mm² (14.4 mm)	10 mm² (4.10 mm) 70 mm² (10.90 mm)	1.97 [50]	0.88 [22]	2.13 [54]	997	U997	U997	U997	2
YGHH-C29C29	3/0 Str. (0.470) 250 kcmil (0.575) 1/2" or 5/8" Rod	3/0 Str. (0.470) 250 kcmil (0.575)	95 mm² (12.5 mm) 120 mm² (14.4 mm)	95 mm² (12.5 mm) 120 mm² (14.4 mm)	2.06 [52]	0.88 [22]	2.13 [54]	997	U997	U997	U997	2
YGHH-C34C26 ⑥	300 kcmil (0.630) 500 kcmil (0.813) 3/4" Rod	6 Sol. (0.162) 2/0 Str. (0.419)	150 mm² (16 mm) 240 mm² (20.35 mm)	10 mm² (4.10 mm) 70 mm² (10.90 mm)	2.42 [62]	0.88 [22]	2.13 [54]	1011	U1011	S1011	P1011	4
YGHH-C34C29 ⑥	300 kcmil (0.630) 500 kcmil (0.813)	3/0 Str. (0.470) 250 kcmil (0.575)	150 mm² (16 mm) 240 mm² (20.35 mm)	95 mm² (12.5 mm) 120 mm² (14.4 mm)	2.67 [68]	0.88 [22]	2.13 [54]	1011	U1011	S1011	P1011	4
YGHH-C34C34 ⑥	300 kcmil (0.630) 500 kcmil (0.813)	300 kcmil (0.630) 500 kcmil (0.813)	150 mm² (16 mm) 240 mm² (20.35 mm)	150 mm² (16 mm) 240 mm² (20.35 mm)	2.91 [74]	1.10 [28]	2.58 [66]	1011	U1011	S1011	P1011	6

② For ground rod to wire applications, ground rod must be precrimped; see table for appropriate precrimp dies

③ Where a "U" or "PU" die is recommended with the Y45 HYPRESS™, a PT6515 adapter must be used

④ Where a "U" or "PU" die is recommended with the Y46 HYPRESS™, a PUADP-1 adapter must be used

⑥ These connectors cannot be installed with the Y35 or Y39 HYPRESS™

Ground Rod Dia.	PRECRIMP Dies	
1/2"	UPRECRIMP 12	U2CABT
5/8"	UPRECRIMP 58	
3/4"	UPRECRIMP 34	

TYPE YGF

GROUNDING PLATE

The irreversible compression ground plate is designed to withstand the rigors of concrete construction. The ground plates are made of high strength, high-conductivity cast copper alloy body with a pure wrought copper compression element. In addition to the tapped NEMA size holes and spacing on the face, the plate comes with a tapped hole on the underside for ease of positioning prior to pouring the concrete. UL467 Listed. Acceptable for direct burial in earth or concrete. Prefilled with PENETROX™ compound and strip sealed.

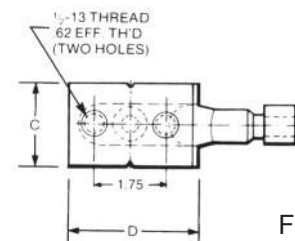


Fig. 1

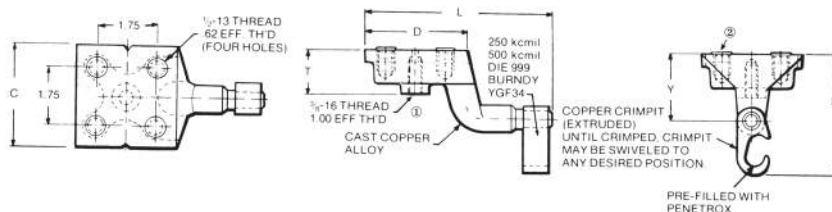


Fig. 2

NOTES:

- ① This tapped hole may be used to position the grounding plate on a threaded rod prior to placement of the concrete. 3/8-16 thread with 1.00 EFF. Thread is standard. If other thread is required, add appropriate suffix code to catalog number. -50 (1/2-13, .94 EFF. Thread), -62 (5/8-11, .94 EFF. Thread) and -75 (3/4-10, .81 EFF. Thread) Example: YGF34-4N-50 is YGF34-4N with 1/2-13 Thread
- ② Plastic plugs are provided to keep dirt out of the threaded holes until the attachment of grounding terminals


IEEE-837*


16060-2.03.C.4

16060-2.03.C.2

Catalog Number	Fig. #	C	D	H	L	T	Y	Copper Conductor Range	Tapped Holes		Installation Tools, Die Set Cat. No., and (# of Crimps)		
									Size	Hole Centers	HYPRESS™		
											PAT750/ Y750/Y35/ Y39	Y45 ①	Y46 ②
YGF29-2N	1	2.00	3.25	3.62	5.78	1.31	2.00	2 AWG-250 kcmil	1/2 - 13	1-3/4	U997 (1)	U997 (1)	U997 (1)
YGF29-4N	2	3.25	3.25	3.62	5.78	1.31	2.00	2 AWG-250 kcmil	1/2 - 13	1-3/4	U997 (1)	U997 (1)	U997 (1)
YGF34-2N ③	1	2.00	3.25	4.62	5.40	1.31	2.19	250 kcmil-500 kcmil	1/2 - 13	1-3/4	U1011 (3)	S1011 (2)	P1011 (2)
YGF34-4N ③	2	3.75	3.75	4.62	5.90	1.31	2.19	250 kcmil-500 kcmil	1/2 - 13	1-3/4	U1011 (3)	S1011 (2)	P1011 (2)

① Where a "U" or "PU" die is recommended with the Y45 HYPRESS™, a PT6515 adapter must be used

② Where a "U" or "PU" die is recommended with the Y46 HYPRESS™, a PUADP-1 adapter must be used

③ These connectors can only be installed using the Y750, Y45 or Y46 HYPRESS™ with recommended dies, they cannot be installed with the Y35 or Y39 HYPRESS™

*Please contact Customer Service for applications requiring IEEE-837 2014.

MECHANICAL GROUNDING CONNECTORS

16060-2.03.C.2

More than 60 years of technological innovation has made BURNDY® mechanical grounding connectors one of the most widely used, highly respected lines in the industry. There is virtually no grounding application problem that this diversified line cannot help solve.

All BURNDY mechanical grounding connectors have been designed for easy installation and for outstanding durability. Only the finest high copper alloys are used in their manufacture, ensuring top performance under the most extreme environmental conditions.

UL467 Listed for direct burial applications in earth or concrete.

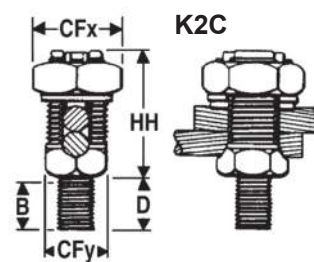
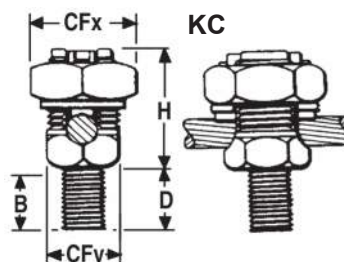
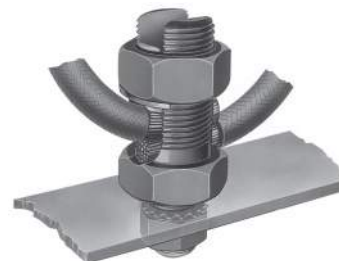
E-32

TYPES KC, K2C

SERVIT POST™

For Copper Cable to Flat

SERVIT POST™ used to ground one or two cables to steel structures, fence posts, transformers. Also used to tap one or two cables from bus bar. One-wrench installation.



TYPE KC One Wire	TYPE K2C One or Two Wires	Stranded	Solid	Stud Diameter	B	CFx	CFy	D	H	HH
KC15	K2C15	12 AWG-9 AWG	12 AWG-8 AWG	1/4-20	3/8	1/2	3/8	1/2	5/8	7/8
KC15B1	K2C15B1	12 AWG-9 AWG	12 AWG-8 AWG	1/4-20	7/8	1/2	3/8	1	5/8	7/8
KC17	K2C17	10 AWG-7 AWG	10 AWG-6 AWG	1/4-20	3/8	5/8	7/16	1/2	7/8	1
KC17B1	K2C17B1	10 AWG-7 AWG	10 AWG-6 AWG	1/4-20	7/8	5/8	7/16	1	7/8	1
KC20	K2C20	10 AWG-5 AWG	10 AWG-4 AWG	5/16-18	13/32	11/16	1/2	5/8	7/8	1-1/8
KC20B1	K2C20B1	10 AWG-5 AWG	10 AWG-4 AWG	5/16-18	27/32	11/16	1/2	1	7/8	1-1/8
KC22	K2C22	10 AWG-3 AWG	10 AWG-2 AWG	3/8-16	15/32	3/4	5/8	5/8	1	1-1/4
KC22B1	K2C22B1	10 AWG-3 AWG	10 AWG-2 AWG	3/8-16	31/32	3/4	5/8	1-1/8	1	1-1/4
KC23	K2C23	8 AWG-2 AWG	10 AWG-1 AWG	3/8-16	15/32	13/16	5/8	5/8	1	1-3/8
KC23B1	K2C23B1	8 AWG-2 AWG	10 AWG-1 AWG	3/8-16	31/32	13/16	5/8	1-1/8	1	1-3/8
KC25	K2C25	2 AWG-1/0 AWG	2 AWG-2/0 AWG	1/2-13	9/16	15/16	3/4	3/4	1-1/8	1-5/8
KC25B1	K2C25B1	2 AWG-1/0 AWG	2 AWG-2/0 AWG	1/2-13	1-1/16	15/16	3/4	1-1/4	1-1/8	1-5/8
KC26	K2C26	2 AWG-2/0 AWG	2 AWG-3/0 AWG	1/2-13	17/32	1	7/8	3/4	1-3/8	1-7/8
KC26B1	K2C26B1	2 AWG-2/0 AWG	2 AWG-3/0 AWG	1/2-13	1-1/16	1	7/8	1-1/4	1-3/8	1-7/8
KC28	K2C28	1 AWG-4/0 AWG	1 AWG-4/0 AWG	5/8-11	3/4	1-1/2	1-3/16	1	1-3/4	2-1/4
KC28B1	K2C28B1	1 AWG-4/0 AWG	1 AWG-4/0 AWG	5/8-11	1-1/4	1-1/2	1-3/16	1-1/2	1-3/4	2-1/4
—	K2C28G3	1 AWG-4/0 AWG	1 AWG-4/0 AWG	1/2-13	1-1/4	1-1/2	1-3/16	1-1/2	1-3/4	2-1/4
KC31	K2C31	1 AWG-350 kcmil	N/A	5/8-11	3/4	1-11/16	1-3/8	1	2-1/4	2-7/8
KC31B1	K2C31B1	1 AWG-350 kcmil	N/A	5/8-11	1-1/4	1-11/16	1-3/8	1-1/2	2-1/4	2-7/8
KC34	K2C34	3/0 AWG-500 kcmil	N/A	3/4-10	1	2	1-5/8	1-1/4	2-3/8	3-1/4
KC34B1	K2C34B1	3/0 AWG-500 kcmil	N/A	3/4-10	1-1/2	2	1-5/8	1-3/4	2-3/8	3-1/4

NOTE:

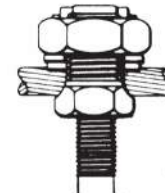
** Not UL 467 rated

Add "-NSP" suffix to catalog numbers to have connector supplied with split lockwasher and nut.

TYPES KC-J12, EQC632C

Transformer Ground Connectors
for Copper

Fits all standard EEI-NEMA distribution transformers as tank grounding terminal.



1/2-13
stud size

Type KC22B2

Catalog Number	Ranges
KC22J12T13	8 Sol. - 2 Sol.
KC26	2 Sol. - 2/0 Str.
KC34J12T13	3/0 - 500 Str.
EQC632C	8 Sol. - 2 Str.

E-35

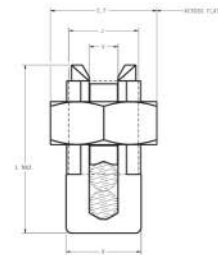
TYPE KS

SERVIT®

16060-2.03.C.2

For Copper

UL467 Listed for direct burial applications in earth or concrete. Compact, highstrength, high copper alloy SERVIT® split-bolt has free-running threads and easy to grip wrench flats. Highly resistant to cracking and corrosion.



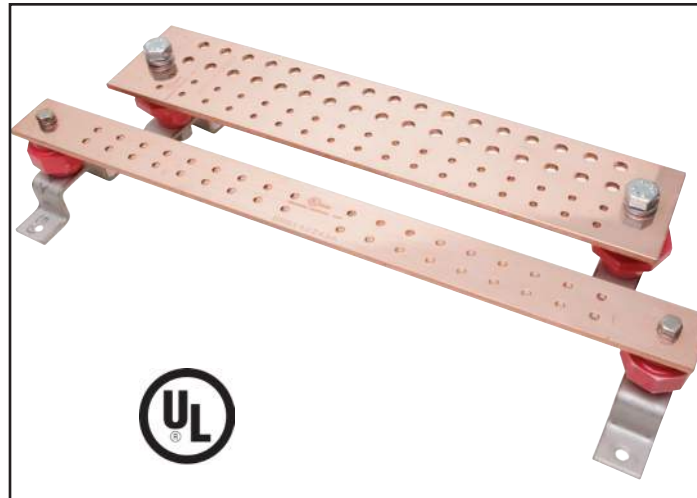
Catalog Number	Cross Flats	L	W	Copper Conductor Range	Rebar with (1) #8 Sol. Cu	Recommended Tightening Torque (in-lb)
KS15	0.50	0.85	0.38	10 - 8 Str.	14 Str.	80
KS17	0.63	1.14	0.45	8 Str. - 6 Sol.	14 Str.	165
KS20	0.69	1.20	0.51	8 Str. - 4 Sol.	14 Str.	165
KS22	0.75	1.50	0.60	6 Str. - 2 Sol.	14 Str.	275
KS23	0.82	1.54	0.62	6 Str. - 2 Str.	14 Str.	275
KS25	0.94	1.77	0.73	4 Str. - 1/0 Str.	14 Str.	385
KS26	1.05	1.94	0.82	2 Str. - 2/0 Str.	14 Str.	385
KS27	1.36	1.86	1.17	1 Str. - 3/0 Str.	8 Sol.	500
KS29	1.36	2.07	1.17	1 Str. - 250	8 Str.	650
KS31	1.70	2.51	1.41	1/0 Str. - 350	1/0 Str.	650
KS34	1.82	2.79	1.48	2/0 Str. - 500	2/0 Str.	825

TYPE BBB

BURNDY® Bus Bar

Copper Bus Bar

Bare copper BusBar, UL Listed for grounding. Available in many sizes and hole patterns. Brackets and insulators included with most styles. Also available in undrilled, horizontal and vertical versions. BusBar is used in a variety of applications. Can be used as a common ground point and "power" applications as well.



Catalog Number	Fig No.	Insulator & Bracket	T - Bar Thickness	W - Bar Width	L - Bar Length	E1	E2	E3	F1	F2	No. of Holes	K
BBB-14210A	1	0.63	1/4"	2 in	10"	0.63	—	—	1.00	—	16	0.28
BBB-14224B	2	0.62	1/4"	2 in	24"	0.62	0.75	—	1.00	1.00	36	0.28
BBB-14410C	3	0.75	1/4"	4 in	10"	0.75	1.00	—	1.25	—	22	0.44
BBB-14410D	4	1.25	1/4"	4 in	10"	1.25	1.00	—	1.13	1.13	22	0.44
BBB-14412E	5	0.75	1/4"	4 in	12"	0.75	1.00	—	2.00	1.25	18	0.44
BBB-14412F	6	1.00	1/4"	4 in	12"	1.00	0.75	—	2.00	1.25	24	0.44
BBB-14416G	7	0.75	1/4"	4 in	16"	0.75	1.00	—	1.69	—	24	0.44
BBB-14416H	8	1.00	1/4"	4 in	16"	1.00	1.00	0.75	1.69	—	32	0.44
BBB-14420J	9	1.00	1/4"	4 in	20"	1.00	1.00	0.75	1.00	—	68	0.44
BBB412UD	—	N/A	1/4"	4 in	12"	N/A	N/A	N/A	N/A	N/A	0	—
BBB-424-UD	—	N/A	1/4"	4 in	24"	N/A	N/A	N/A	N/A	N/A	0	—
BBB-HR19	—	0.38	3/16"	3/4 in	19"	0.38	—	—	—	—	8	—
BBB-VR36	—	0.32	1/4"	5/8 in	36"	0.32	—	—	—	—	16	—

* Contact factory for custom sizes.

NOTE:

For separately ordering Insulator & Brackets use catalog number **B38723000**.

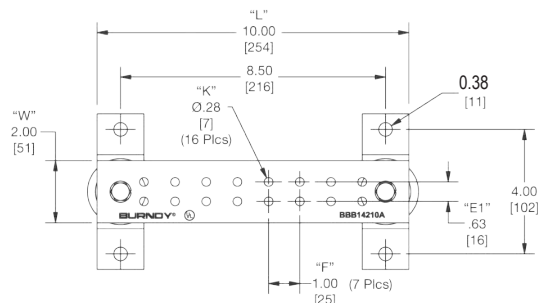


Figure 1

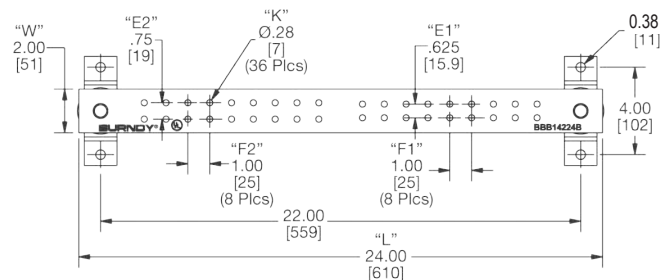


Figure 2

BURNDYWeld®
QIKLITE®**BURNDYWeld® Weld Metal****Battery Ignition System**

The self-contained unit offers a built-in battery life indicator, 6' ignitor cord and separate buttons for power and ignition. Also, no starting powder is used with this system so emissions are greatly reduced. QIKLITE® works with "off-the-shelf" molds and weld metal, so contractors and installers can continue to use the same quality BURNDYWeld® products.

Since the unit operates with standard weld metal and molds, installers also have the option of using a traditional flint ignitor or the new QIKLITE® system. Another feature is the speed of ignition, with virtually no delay between depressing the "push to operate" button and ignition of the weld metal. This instant ignition feature offers quick reassurance to the installer, without the delay.



E-83

Features & Benefits

- The QIKLITE® system allows installers to make exothermic connections remotely
- Uses standard exothermic molds and weld metal
- Use the new BURNDYWeld® QIKLITE® system or use a flint ignitor, this versatility makes this system better than anything available in the industry
- Installers have virtually 100% confidence that the connection can be completed with no wasted material or weld metal shots
- No starting material required - which greatly reduces the amount of emissions generated
- Power is supplied by 4 standard "D" Size batteries
- Built-in battery life indicator
- There's virtually no delay in ignition after depressing the "push to operate" button
- Separate ON/OFF and Operate buttons
- Comes complete with durable 6' heat resistant cord
- Durable, long-lasting design
- QIKSTIK ignitor sticks are conveniently packaged and available separately or available packaged complete with standard weld metal

Catalog Number Standard Cartridge Size with Ignitor	Cartridges/ Ignitors Per Box
15Q	20
25Q	20
32Q	10
45Q	20
65Q	20
90Q	10
115Q	10
150Q	10
200Q	10
250Q	10
500Q	10
QIKSTIK*	10

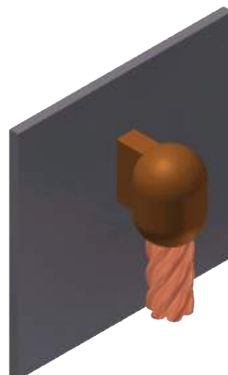
* Ignitor stick only.

Catalog Number	Description
QIKLITE	Battery Ignition System
QIKLITEKIT	QIKLITE® Battery Ignition System and 20 QIKSTIK Ignitor Sticks

TYPE BCS-23 MOLDS

Vertical Cable Drop to Vertical Steel Surface

Type BCS-23 Molds are used to join vertical cable down to a vertical steel surface. Note that the cable is OFF the surface. Size range is #6 through 500 kcmil solid or concentric stranded copper conductors. Contact BURNDY® for information on molds for conductors not listed below.



NOTES:

- For sizes not listed, contact BURNDY®
- Molds listed are for concentric stranded cable; add suffix "-S" to mold number for solid conductors
- For heavy duty molds, molds with wear plates or molds for Copperweld® cable, contact BURNDY®
- For expedited service, contact BURNDY®

Required Tools:

Handle Clamps - see chart for correct handle catalog number

Recommended Accessories:

B38-0309-00 Flint Ignitor; **B38-3922-00** Mold Cleaning Brush; **B38-0135-00** Cable Cleaning Brush; **B38-0330-00** Cable Clamp

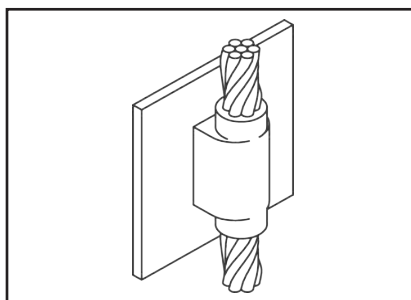
E-100

Cable Size	Mold Number	Price Key	Weld Metal	Handle Clamps
#6	B-5389	4	45	B-106
#4	B-5359	4	65	B-106
#2	B-2781	4	65	B-106
#1	B-5361	4	90	B-106
1/0	B-2189	4	115	B-106
2/0	B-2540	4	115	B-106
3/0	B-5362	4	150	B-106
4/0	B-8718	4	150	B-106
250 kcmil	B-8165	4	200	B-106
300 kcmil	B-5363	4	200	B-106
350 kcmil	B-9029	4	250	B-106
500 kcmil	B-8512	17	2-150	B-106

TYPE BCS-4 MOLDS

Vertical Through Cable to Vertical Steel Surface

Type BCS-4 Molds are used to join a vertical through copper cable to a vertical steel surface. Note that the cable is OFF the surface. For molds with the cable on the surface, contact BURNDY®. Size range is #6 through 500 kcmil solid or concentric stranded copper conductors. Contact BURNDY® for information on molds for conductors not listed.



NOTES:

- For sizes not listed, contact BURNDY®
- Molds listed are for concentric stranded cable; add suffix "-S" to mold number for solid conductors
- For heavy duty molds, molds with wear plates or molds for Copperweld® cable, contact BURNDY®
- For expedited service, contact BURNDY®

Required Tools:

Handle Clamps - see chart for correct handle catalog number

Recommended Accessories:

B38-0309-00 Flint Ignitor; **B38-3922-00** Mold Cleaning Brush; **B38-0135-00** Cable Cleaning Brush; **B38-0330-00** Cable Clamp

Cable Size	Mold Number	Price Key	Weld Metal	Handle Clamps
#6	B-1215	4	90	B-106
#4	B-1216	4	90	B-106
#2	B-1218	4	115	B-106
#1	B-1219	4	115	B-106
1/0	B-1220	17	200	B-106
2/0	B-1221	17	200	B-106
3/0	B-1222	17	250	B-106
4/0	B-1223	17	250	B-106
250 kcmil	B-1224	17	250	B-106
300 kcmil	B-1225-H	6	500	B-107
350 kcmil	B-1226-H	6	3-200	B-107
500 kcmil	B-1228-H	6	3-250	B-107

2.0 GROUND RODS

SPECIFICATION – 16060–2.02.B GROUNDING AND BONDING



REQUEST FOR INFORMATION (RFI) RESPONSE

Owner:	<u>County of Hawai'i</u>	Project No.:	<u>203795</u>
Contractor:	<u>Nan, Inc.</u>	RFI No.:	<u>0039</u>
Project Name:	<u>Hilo WWTP Rehabilitation and Replacement Project Phase 1</u>	Spec/Dwg Reference:	<u>16060-2.03.A</u>
RFI Title:	<u>Ground Rod Type</u>		

Information Requested	
MECI requests that a 0.75-inch, 10-foot pointed ground rod be used in lieu of the 0.75-inch, 10-foot threaded ground rod with threaded coupling and steel-driving stud detailed in specification 16060.2.03.A. MECI is intending to drive ground rods utilizing the iTOOL ROD DAWG XL which does not require a steel-driving stud. This method has proven to enhance productivity and mitigate safety concerns from eliminating the need to work from heights. Furthermore, should supplemental ground rods be required, non-threaded rods from approved manufactures (Erico) are widely available from local vendors on island.	
Authored By:	Date Submitted:
<u>Nan, Inc.</u>	<u>06/05/2025</u>

Response to Request:	
The pointed ground rods are acceptable.	
In the event, Contractor believes the RFI response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm Name:	Date Returned:
<u>Carollo Engineers</u>	<u>6/12/2025</u>
Signature:	
<u>Kellen Gross</u>	

END OF DOCUMENT

REQUEST FOR INFORMATION

PROJECT NO: WW-4705R
PROJECT: Hilo WWTP Rehabilitation and Replacement Phase 1

NAN RFI No. 39
Sub RFI No. 024
Bid Item No. N/A
NAN Proj No. 24-077

OWNER: County of Hawaii, Department of Environmental
Management, Wastewater Division

CONTRACTOR: Nan, Inc.

SUBCONTRACTOR: Mass Electric Construction Co.

			YES	NO	UNKNOWN
Date Submitted to Owner:	Thursday, June 5, 2025	POTENTIAL COST IMPACT:	☐	☒	☐
Response Requested NLT:	Thursday, June 19, 2025	POTENTIAL TIME IMPACT:	☐	☒	☐

RFI Title:

RFI Type:

☐ Confirmation of Contract Requirements	☐ Request for Clarification	☒ Request for Variance	☐ Other _____
☐ Conflict in Drawings / Specification	☐ Differing Site Condition	☐ Design Error/Omission	

Referenced Specifications:
16060-2.03.A

Referenced Drawings:
MULTIPLE

INFORMATION REQUESTED:

MECI requests that a 0.75-inch, 10-foot pointed ground rod be used in lieu of the 0.75-inch, 10-foot threaded ground rod with threaded coupling and steel-driving stud detailed in specification 16060.2.03.A. MECI is intending to drive ground rods utilizing the iTOOL ROD DAWG XL which does not require a steel-driving stud. This method has proven to enhance productivity and mitigate safety concerns from eliminating the need to work from heights. Furthermore, should supplemental ground rods be required, non-threaded rods from approved manufactures (Erico) are widely available from local vendors on island.

Proposed by: Mass Electric Construction Co. Date: June 5, 2025

PROPOSED SOLUTION:

Proposed by: Mass Electric Construction Co. Date: June 5, 2025

RESPONSE:

Responded by: _____ Date: _____

COPPER-BONDED GROUND ROD, POINTED, 3/4" DIA, 10', 10 MIL PLATING, 12.6 LB

CATALOG NUMBER

613400



CERTIFICATIONS



FEATURES

- 99.9% pure electrolytic copper coating 16060-2.03.A.2
- Rods have a high carbon steel core and tip that provide superior strength when driving
- Copper coating will not crack when bent or tear when driven
- Minimum copper coating of 10 mils on rods listed to UL® 467 16060-2.03.A.3
- nVent ERICO name, length, diameter and part number is roll-stamped within 12" (304,8 mm) of chamfered end
- UL logo and control number where applicable stamped on each rod for easy inspection after installation
- Molecular bond to nickel-sealed high strength steel core

PRODUCT ATTRIBUTES

- Material: Copper-Bonded Steel
- Tensile Strength: 80000 lbf/in² Min
- Ground Rod Diameter, Nominal: 3/4" 16060-2.03.A.1
- Ground Rod Diameter, Actual (Ø): 0.68"
- Length (L): 10' 16060-2.03.A.1

Plating Thickness: 10mil

Unit Weight: 12.6 lb

UPC Label: No

Complies With: IEC® EN 62561-2;ANSI®/NEMA® GR1

ADDITIONAL PRODUCT DETAILS

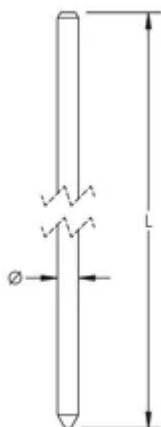
For rods to be listed to UL® 467, they must be at least 8' (2.44 m) in length.

16060-2.03.A.3

IEC® EN 62561-2 supercedes EN 50164-2.

Additional lengths available.

DIAGRAMS



WARNING

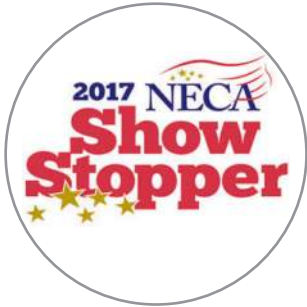
nVent products shall be installed and used only as indicated in nVent's product instruction sheets and training materials. Instruction sheets are available at www.nvent.com and from your nVent customer service representative. Improper installation, misuse, misapplication or other failure to completely follow nVent's instructions and warnings may cause product malfunction, property damage, serious bodily injury and death and/or void your warranty.

⚠ WARNING: This product can expose you to chemicals including nickel, which is known to the State of California to cause cancer and birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov.

GROUND ROD DAWG™

DRIVE GROUND RODS SAFELY IN SECONDS

iTOOLco.



Safely drive 1/2", 3/8", 5/8", and 3/4" rods from the ground in under a minute with iTOOLco's Ground Rod Dawg™. Once rod is driven down far enough the tool will hammer from top to finish driving rod under grade levels from ground.

“WE HAD A LARGE DATA CENTER THAT REQUIRED GROUND RODS TO BE DRIVEN ALL AROUND THE BUILDING. The iTOOLco Ground Rod Dawg™ was a real time saver and provided safety we never knew existed. We used to drag a ladder around and drive rods from the top. Not anymore! Thanks for another fantastic tool iTOOLco.

Gordon Gebman
Foreman
Omega Electric



Safely drive rods from ground



Tool locks into hammer drill



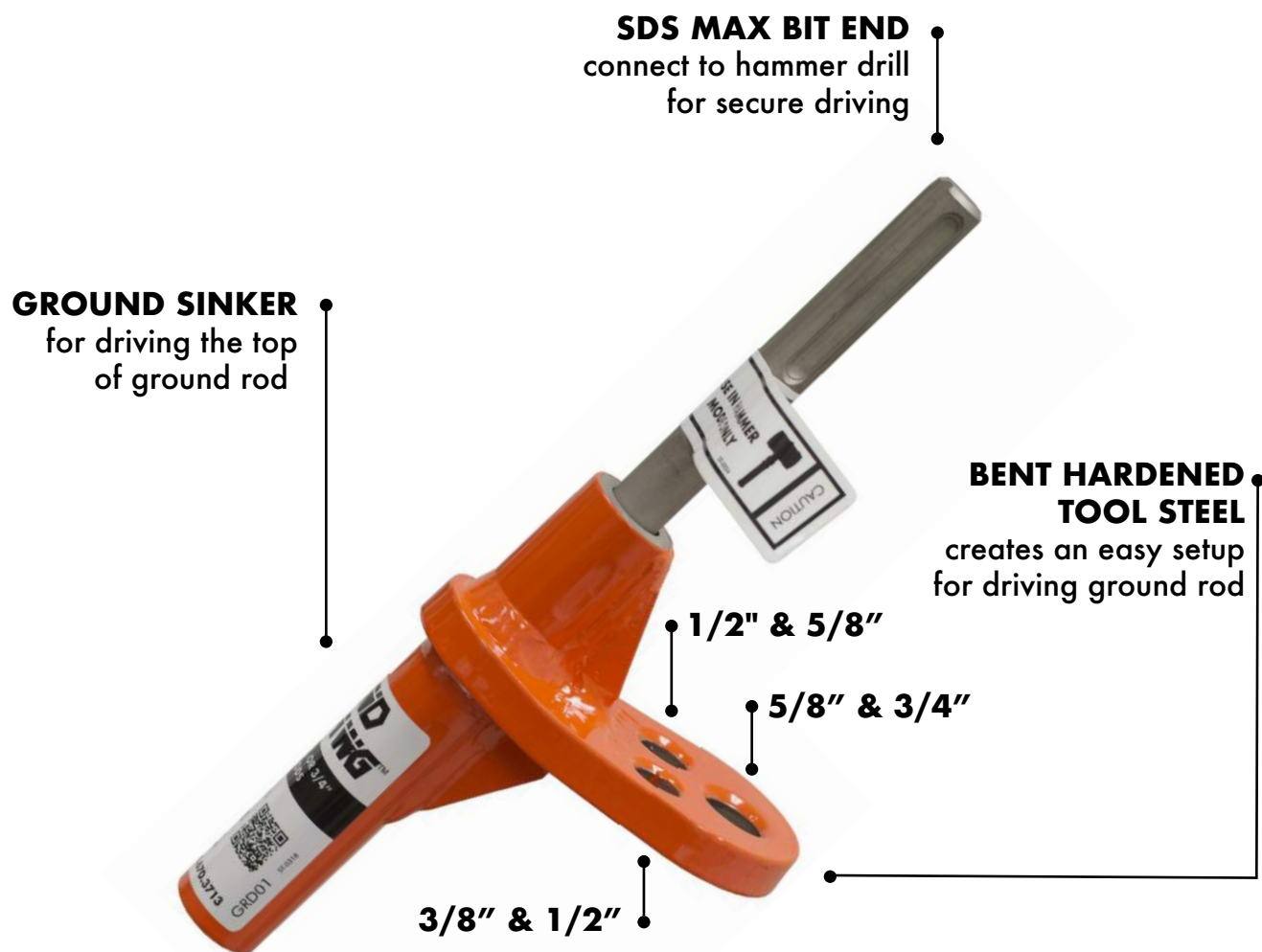
Fits 1/2", 5/8" and 3/4"

FASTER SAFER WORKSPACE

GROUND ROD DAWG™

DRIVE RODS FASTER AND EASIER THAN EVER

iTOOLco.



Model	Description	Specifications
GRD01	Ground Rod Dawg™	11-3/4"H x 2-3/4"W x 5"L 4 lb.
UPC #650270083458		

6
YEAR
LIMITED
WARRANTY

MADE IN THE
USA
WITH GLOBAL
COMPONENTS

Patent Pending

GROUND ROD DAWG XL™

DRIVE LARGE ROD FROM GROUND LEVEL

iTOOLco.



With two models available, safely drive three quarter or one inch rods from ground level with iTOOLco's Ground Rod Dawg XL's. The Ground Rod Dawg XL eliminates the need for climbing up ladders and using sledge hammers to drive large ground rods under grade. Simply plug the Ground Rod Dawg XL into your hex shank jack hammer. Then insert the bottom of the rod into the Ground Rod Dawg XL and go. The unique design allows the Ground rod Dawg XL to bite down onto the rod, allowing the user to drive it in with ease. Once the rod is driven in far enough, use the end receiver on the top of the rod to finish driving it to the final grade level.



No Ladder Needed to Drive Rod



End Receiver Drives Top of Rod Below Grade



Fits in 1-1/8" Hex Shank Jack Hammer

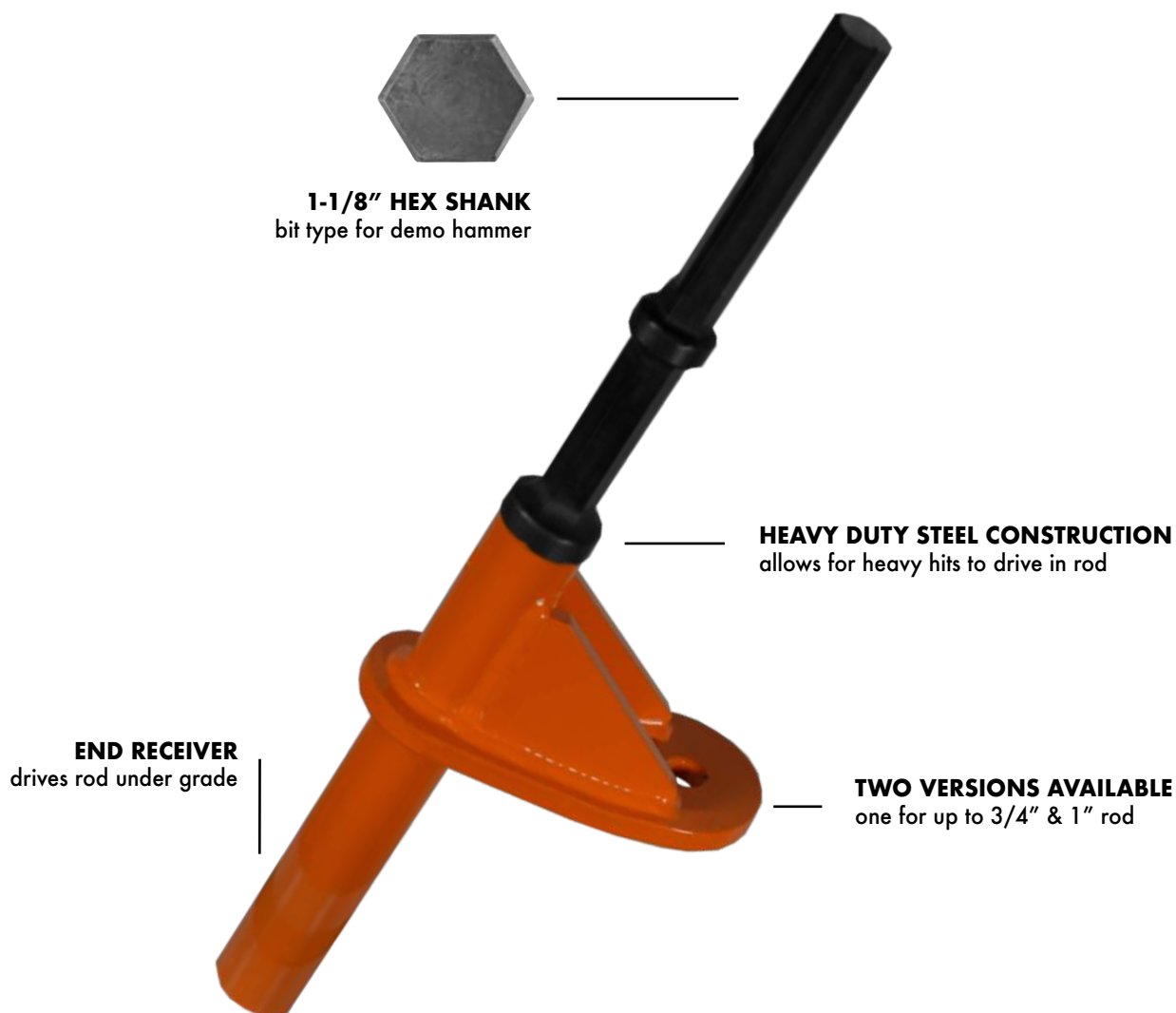
FASTER SAFER WORKSPACE

iTOOLco.com / 865-670-3713

GROUND ROD DAWG XL™

DRIVE LARGE ROD FROM GROUND LEVEL

iTOOLco.



*Sold as shown

Model	Description	Specifications
GRDXL34	Ground Rod Dawg XL™ (Up to 3/4")	4"H x 9-1/2"W x 23"L 18 lb.
GRDXL01	Ground Rod Dawg XL™ (Up to 1")	4"H x 9-1/2"W x 23"L 18 lb.

6
YEAR
LIMITED
WARRANTY

Patent Pending

3.0 GROUND CABLES

SPECIFICATION – 16060-2.02.C GROUNDING AND BONDING





Bare Copper Solid, Compressed Class B and C



Image not to scale. See Table 1 for dimensions.

CONSTRUCTION:

- Conductor:** Bare copper solid or compressed class B or C per ASTM B3 and **ASTM B8** — 16060-2.03.B.1.b

SPECIFICATIONS:

- **ASTM B3 Soft or Annealed Copper Wire** — 16060-2.03.B.1.a
- **ASTM B8 Concentric-Lay-Stranded Copper Conductors** — 16060-2.03.B.1.b

Table 1 – Physical and Electrical Data

Stock Number	Cond. Size	Strand Class	Strand Count	Cond. Cmil	Cond. Number	Diameter Over Conductor	Approx. OD	Approx. Weight	DC Resistance @ 25°C	AC Resistance @ 75°C	Rated Strength
	AWG/kcmil		No. of Strands	cmil	No.	inch	inch	lb/1000ft	Ω/1000ft	Ω/1000ft	lb
207746 [◇]	2/0	B	19	133100	1	0.405	0.405	410	0.081	0.097	1064

All dimensions are nominal and subject to normal manufacturing tolerances

◇ Cable marked with this symbol is a standard stock item

TBA stock codes are estimations only and actual product may vary. Please wait until a stock code is assigned to purchase connectors and/or fittings.





Bare Copper Solid, Compressed Class B and C



Image not to scale. See Table 1 for dimensions.

CONSTRUCTION:

- Conductor:** Bare copper solid or compressed class B or C per ASTM B3 and **ASTM B8** — 16060-2.03.B.1.b

SPECIFICATIONS:

- **ASTM B3 Soft or Annealed Copper Wire** — 16060-2.03.B.1.a
- **ASTM B8 Concentric-Lay-Stranded Copper Conductors** — 16060-2.03.B.1.b

Table 1 – Physical and Electrical Data

Stock Number	Cond. Size	Strand Class	Strand Count	Cond. Cmil	Cond. Number	Diameter Over Conductor	Approx. OD	Approx. Weight	DC Resistance @ 25°C	AC Resistance @ 75°C	Rated Strength
	AWG/kcmil		No. of Strands	cmil	No.	inch	inch	lb/1000ft	Ω/1000ft	Ω/1000ft	lb
427633 [◇]	4/0	B	19	211600	1	0.512	0.512	653	0.051	0.062	1692

All dimensions are nominal and subject to normal manufacturing tolerances

◇ Cable marked with this symbol is a standard stock item

TBA stock codes are estimations only and actual product may vary. Please wait until a stock code is assigned to purchase connectors and/or fittings.





XHHW-2 Copper Circuit Sizes (14, 12 & 10 AWG) Silicone Free

Power Cable 600 or 1000 Volt Single Conductor Copper, Cross Linked Polyethylene (XLPE). Silicone Free

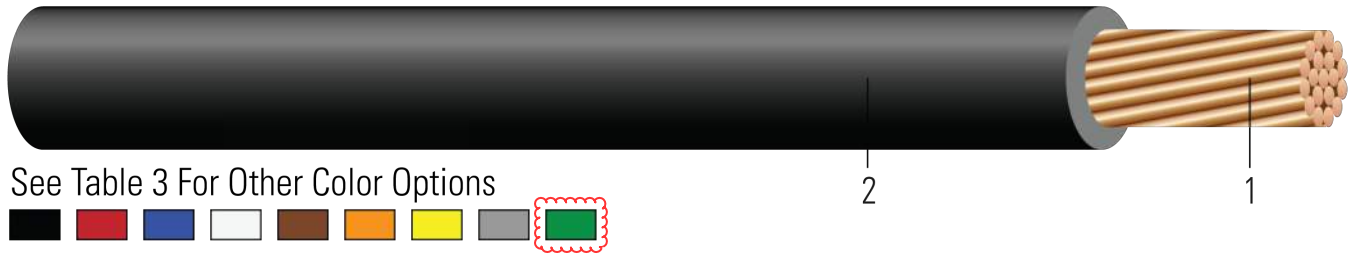


Image not to scale. See Table 1 for dimensions.

CONSTRUCTION: 16060-2.03.B.2.a

1. **Conductor:** Solid or Class B compressed stranded bare copper per ASTM B3 and ASTM B8
2. **Insulation:** Cross Linked Polyethylene (XLPE)

APPLICATIONS AND FEATURES:

APPLICATION

Southwire XHHW-2 copper conductors circuit sizes are primarily used in conduit, or other recognized raceways, and branch circuit wiring, as specified in the National Electrical Code. XHHW-2 copper conductors may be used in wet or dry locations at temperatures not to exceed 90°C. Voltage rating for XHHW-2 conductors is 600 volts or 1000 volts. Suitable for use in Health Care Facilities per Section 517.160 of the National Electrical Code where a dielectric constant of less than 3.5 maybe specified.

FEATURES

- SIS - Sizes 14 AWG through 10 AWG
- RoHS Compliant

SPECIFICATIONS:

- ASTM B3 Soft or Annealed Copper Wire
- ASTM B8 Concentric-Lay-Stranded Copper Conductors
- UL 44 Thermoset-Insulated Wires and Cables
- ICEA S-95-658 (NEMA WC70) Power Cables Rated 2000 Volts or Less for the Distribution of Electrical Energy
- Made in America: Compliant with both Buy American and Buy America Act (BAA) requirements per 49 U.S.C. § 5323(j) and the Federal Transit Administration Buy America requirements per 49 C.F.R. part 661
- Federal Specification A-A-59544
- NMX-J-451-ANCE Thermoset insulated wires and cables
- NOM-063-SCFI Electrical Products – Conductors – Safety Requirements
- NEMA 70901-2-2024 Make It American Compliance with Domestic Preference Requirements Pt. 2 Wire & Cable

SAMPLE PRINT LEGEND:

14 AWG thru 10 AWG

SOUTHWIRE E30117 (UL) XX AWG (XXXmm²) CU TYPE XHHW-2 600V/1000V OR SIS 600 VOLTS NOM-ANCE




Table 1 – Weights and Measurements

Cond. Size	Strand Count	Diameter Over Conductor	Insul. Thickness	Insulation Color	Approx. OD	Copper Weight	Approx. Weight
AWG/Kcmil	No. of Strands	inch	mil		inch	lb/1000ft	lb/1000ft
14	7	0.070	30	PE	0.131	12	17
12	Solid	0.080	30	BK	0.141	19	25
12	7	0.088	30	PE	0.147	20	24
10	Solid	0.101	30	BK	0.162	30	37
10	7	0.113	30	BK	0.173	32	36

All dimensions are nominal and subject to normal manufacturing tolerances

◊ Cable marked with this symbol is a standard stock item

TBA stock codes are estimations only and actual product may vary. Please wait until a stock code is assigned to purchase connectors and/or fittings.

Table 2 – Electrical and Engineering Data

Cond. Size	Min Bending Radius	Max Pull Tension	DC Resistance @ 25°C	AC Resistance @ 75°C	Inductive Reactance @ 60Hz	Allowable Ampacity At 75°C	Allowable Ampacity At 90°C
AWG/Kcmil	inch	lb	Ω/1000ft	Ω/1000ft	Ω/1000ft	Amp	Amp
14	0.5	32	2.631	3.170	0.058	20	25
12	0.6	52	1.662	2.002	0.054	25	30
12	0.6	52	1.662	2.002	0.054	25	30
10	0.6	83	1.040	1.253	0.050	35	40
10	0.7	83	1.040	1.253	0.050	35	40

* Ampacities based upon 2023 NEC Table 310.16 and do not take into account the overcurrent protection limitations in NEC 240.4(D) of 15 Amps for 14 AWG CU, 20 Amps for 12 AWG CU, and 30 Amps for 10 AWG CU (independent of the conductor temperature rating and stranding if size is present in table). Also, see NEC sections 310.15 and 110.14(C) for additional requirements.

* Inductive Reactance is based on non-ferrous conduit with one diameter spacing center-to-center.

Table 3 - Stock Code Colors

Size (Strand)	Black	Red	Blue	White	Brown	Orange	Yellow	Gray	Purple	Green
12 (Solid)	550220			550221						550223
10 (Solid)	137240	550213	550215	550212						550214
14 (7)	112920	370932	370940	370924	370981	370973	370957	371005	370999	370965
12 (7)	112938	371039	371047	371021	371088	371070	371054	371104	371096	371062
10 (7)	112946	371138	371146	371120	371187	371179	371153	371203		371161

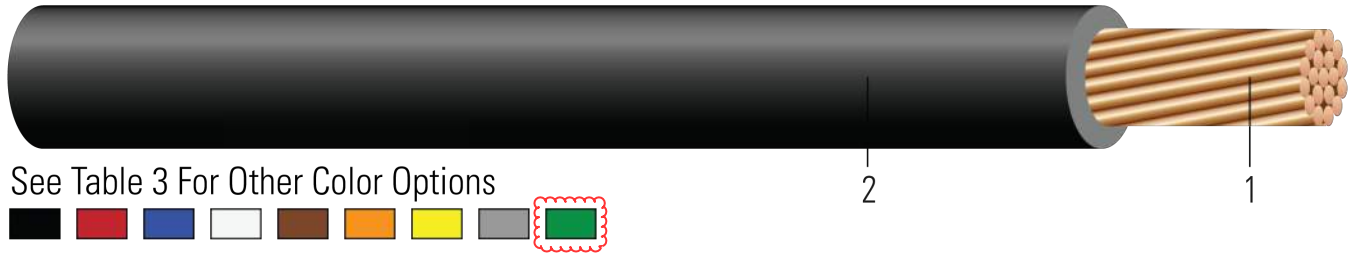
Award Winning Patent
 Pending Building Wire
 Selector





SIMPull XHHW-2[®]/RW90 Copper

Single Conductor Copper, Cross Linked Polyethylene (XLPE) with SIMpull technology for easier pulling. Type XHHW-2 Conductor sizes 8 AWG and larger are rated 600 or 1000 Volt. Type RW90 Conductors are rated for 600 Volt.



See Table 3 For Other Color Options



Image not to scale. See Table 1 for dimensions.

CONSTRUCTION:

1. **Conductor:** Class B compressed stranded bare copper per ASTM B3 and ASTM B8
2. **Insulation:** Cross Linked Polyethylene (XLPE) with SIMpull[®] Technology. Silicone-Free, Abrasion, High-Heat, Moisture Resistant

APPLICATIONS AND FEATURES:

APPLICATION

Southwire Copper SIMpull XHHW-2[®]/RW90 conductors are primarily used in conduit, cable tray or other recognized raceways for services, feeders, and branch circuit wiring. SIMpull XHHW-2[®]/RW90 copper conductors may be used in wet or dry locations at temperatures not to exceed 90°C. Southwire Copper SIMpull XHHW-2[®]/RW90 conductors are designed to be installed without the application of pulling lubricant.

Voltage rating for XHHW-2 is 600 volts or 1000 volts for sizes 8 and larger. Type XHHW-2 is suitable for use in Health Care Facilities per Section 517.160 of the National Electrical Code where a dielectric constant of less than 3.5 maybe specified.

Voltage rating for RW90 is 600 volts. RW90 is suitable for use in open wiring and use in raceways (except cable troughs and ventilated flexible cableways) in dry or wet locations as per Canadian Electrical Code. For open wiring exposed to the weather.

FEATURES

- SIS- 8 AWG
- Sunlight resistant
- -40°C Cold bend
- FT1
- Gasoline and Oil Resistant II
- CT Rated- 1/0 AWG and larger
- FT4- 350 kcmil and larger
- RoHS/REACH Compliant
- SPEC 10005 for circuit sizes 14 AWG, 12 AWG, and 10 AWG

SPECIFICATIONS:

- ASTM B3 Soft or Annealed Copper Wire





- ASTM B8 Concentric-Lay-Stranded Copper Conductors
- ASTM B787 19 Wire Combination Unilay-Stranded Copper Conductors
- UL 44 Thermoset-Insulated Wires and Cables
- UL 1685 Vertical-Tray Fire Propagation and Smoke Release Test (1/0 and Larger)
- CSA C22.2 No. 38 Thermoset-insulated wires and cables
- ICEA S-95-658 (NEMA WC70) Power Cables Rated 2000 Volts or Less for the Distribution of Electrical Energy
- IEEE 1202/FT4 Flame Test (70,000 BTU/hr) 350kcmil and Larger
- Made in America: Compliant with both Buy American and Buy America Act (BAA) requirements per 49 U.S.C. § 5323(j) and the Federal Transit Administration Buy America requirements per 49 C.F.R. part 661
- Federal Specification A-A-59544
- NMX-J-451-ANCE Thermoset insulated wires and cables
- NOM-063-SCFI Electrical Products – Conductors – Safety Requirements
- NEMA 70901-2-2024 Make It American Compliance with Domestic Preference Requirements Pt. 2 Wire & Cable

SAMPLE PRINT LEGEND:**8 AWG thru 1 AWG**

{SQFTG} SOUTHWIRE{R} {NOLUBE}{R} {SIMPULL}{R} E30117 {UL} TYPE XHHW-2 8 AWG (8.37{MM2}) CU 600V/1000V SR GRII PRII OR SIS 600V - LL90458 {CSA} RW90 XLPE 8 AWG (8.37{mm2}) CU 600V GRI PRI -40{D}C SR FT1 - {NOM}-ANCE LS - PAT WWW.PATENTSW.COM

1/0 AWG thru 300 kcmil

{SQFTG} SOUTHWIRE{R} {NOLUBE}{R} {SIMPULL}{R} E30117 {UL} TYPE XHHW-2 1/0 AWG (53.5{MM2}) CU 600V/1000V SR FOR CT USE GRII PRII - LL90458 {CSA} RW90 XLPE 1/0 AWG (53.5{mm2}) CU 600V GRI PRI -40{D}C SR FT1 - {NOM}-ANCE LS - PAT WWW.PATENTSW.COM

350 kcmil and Larger

{SQFTG} SOUTHWIRE{R} {NOLUBE}{R} {SIMPULL}{R} E30117 {UL} TYPE XHHW-2 350 KCMIL (177{MM2}) CU 600V/1000V SR FOR CT USE GRII PRII FT4 - LL90458 {CSA} RW90 XLPE 350 KCMIL (177{mm2}) CU 600V GRI PRI -40{D}C SR FT4 - {NOM}-ANCE LS - PAT WWW.PATENTSW.COM





Table 1 – Weights and Measurements

Cond. Size AWG/Kcmil	Strand Count No. of Strands	Diameter Over Conductor inch	Insul. Thickness mil	Insulation Color	Approx. OD inch	Copper Weight lb/1000ft	Approx. Weight lb/1000ft
8	7	0.141	45	RD	0.238	50	64
6	7	0.177	45	BK	0.274	81	97
4	7	0.225	45	BK	0.321	128	148
3	7	0.252	45	BK	0.348	162	184
2	7	0.282	45	BK	0.379	204	228
1	19	0.322	55	OE	0.435	258	291
1/0	19	0.361	55	PE	0.476	326	361
2/0	19	0.405	55	PE	0.520	410	450
3/0	19	0.456	55	PE	0.570	518	561
4/0	19	0.512	55	PE	0.626	655	704
250	37	0.558	65	PE	0.696	771	834
300	37	0.610	65	GN	0.732	926	992
350	37	0.661	65	PE	0.799	1080	1154
400	37	0.705	65	BE	0.848	1235	1317
500	37	0.789	65	BE	0.908	1543	1632
600	61	0.865	80	PE	1.008	1852	1964
700	61	0.935	80	BN	1.103	2161	2284
750	61	0.968	80	BE	1.107	2315	2439
1000	61	1.117	80	OE	1.285	3087	3236

All dimensions are nominal and subject to normal manufacturing tolerances

◊ Cable marked with this symbol is a standard stock item





Table 2 – Electrical and Engineering Data

Cond. Size	Min Bending Radius	Max Pull Tension	DC Resistance @ 25°C	AC Resistance @ 75°C	Inductive Reactance @ 60Hz	Allowable Ampacity At 75°C	Allowable Ampacity At 90°C
AWG/Kcmil	inch	lb	Ω/1000ft	Ω/1000ft	Ω/1000ft	Amp	Amp
8	1.0	132	0.653	0.786	0.052	50	55
6	1.1	209	0.411	0.495	0.051	65	75
4	1.3	333	0.258	0.310	0.048	85	95
3	1.4	420	0.205	0.246	0.047	100	115
2	1.5	530	0.162	0.195	0.045	115	130
1	1.7	669	0.128	0.154	0.046	130	145
1/0	1.9	844	0.102	0.122	0.044	150	170
2/0	2.1	1064	0.081	0.097	0.043	175	195
3/0	2.3	1342	0.064	0.078	0.042	200	225
4/0	2.5	1692	0.051	0.062	0.041	230	260
250	2.8	2000	0.043	0.053	0.041	255	290
300	2.9	2400	0.036	0.045	0.041	285	320
350	3.2	2800	0.031	0.039	0.040	310	350
400	3.4	3200	0.027	0.035	0.040	335	380
500	3.6	4000	0.022	0.029	0.039	380	430
600	5.0	4800	0.018	0.025	0.039	420	475
700	5.5	5600	0.015	0.022	0.038	460	520
750	5.5	6000	0.014	0.022	0.038	475	535
1000	6.4	8000	0.011	0.018	0.037	545	615

* Ampacities based upon 2023 NEC Table 310.16 and do not consider the overcurrent protection limitations in NEC 240.4(D) of 15 Amps for 14 AWG CU, 20 Amps for 12 AWG CU, and 30 Amps for 10 AWG CU (independent of the conductor temperature rating and stranding if size is present in table). Also, see NEC sections 310.15 and 110.14(C) for additional requirements.

* Inductive impedance is based on non-ferrous conduit with one diameter spacing center-to-center.

* Ampacities derived from the 2021 Canadian Electrical Code. - Table 2 - for Raceway or Cable. Not more than 3 copper conductors on an ambient temperature of 30°C.





Table 3 - Stock Code Colors

Size (Strand)	Black	Red	Blue	White	Brown	Orange	Yellow	Gray	Purple	Green
8 (7)	112953	952713	553059	952721	553060	553061	553062	553063		952739
6 (7)	112961	952705	959916	678607	683383	683391	553067	553068		553230
4 (7)	112979	952697	553846	678599	553847	553848	553849	553850		558627
3 (7)	267278	652971	652972	652973	677646	677647	652975	677648		890469
2 (7)	112987	218115	553087	218107	553088	553089	553090		674066	474122
1 (19)	112995	550761	550762	550808	553854	553855	553856	553857	674065	550766
1/0 (19)	113001	553860	553861	553858	553863	553864	553865	553866	674064	553862
2/0 (19)	113019	553871	553872	553870	553873	553874	553876	553877	553878	552070
3/0 (19)	113027	553881	553882	553880	553885	553886	553887	553888	674063	553884
4/0 (19)	113035	553078	553079	553077	553080	553082	553083	553084	674062	552071
250 (37)	113043	553893	553894	553892	553896	553897	553898	553899	674061	553895
300 (37)	113050	643848	643849	643850	561129	561130	561131	584039	139410	580121
350 (37)	113068	553903	553904	553902	553906	553907	553908	553910	674060	553905
400 (37)	113076	561701	561702	561132	561111	561112	561113	561703		558666
500 (37)	113084	550369	550261	553071	550260	550262	550259	553074	553075	553072
600 (61)	113092	553914	553915	553913	553918	553919	553920	553921	589214	553916
700 (61)	586272				662840	662842	662843	662844		
750 (61)	113100	553927	553928	553926	553930	553931	553932	553934		553929
1000 (61)	113134	138623	138622	138624	678463	678465	678464			668449

Award Winning Patent
Pending Building Wire
Selector



4.0 PRECAST GROUND WELL BOXES

SPECIFICATION – 16060-2.02.D GROUNDING AND BONDING



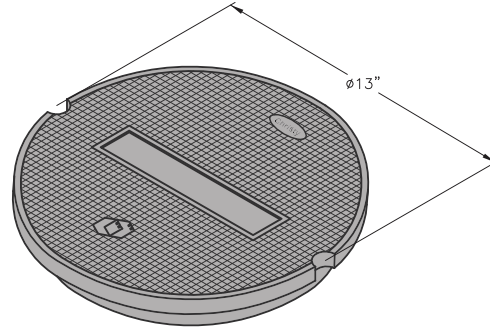
G12

CHRISTY®

COVER

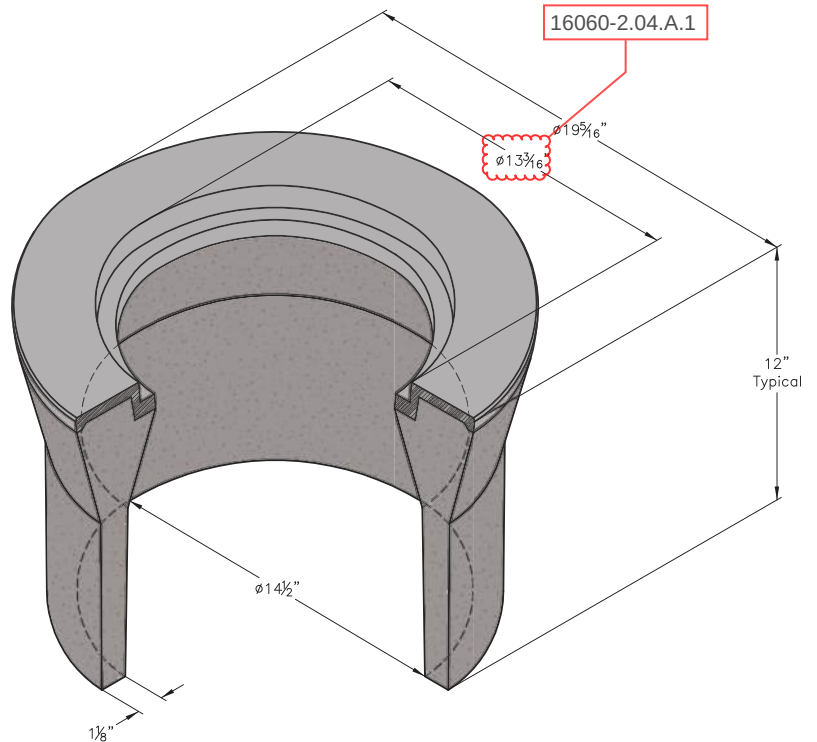
Style:	Flush
Material:	Cast Iron
Model:	13" Dia.
Weight:	24 lbs
Options:	Special Markings
Surface:	Skid Resistant & Marked*
Coefficient of Friction:	>0.6 ASTM 1028
Performance:	H20, AASHTO

Cover comes standard with permanent markings for manufacturer, load rating, model size and manufacturing location.



BODY

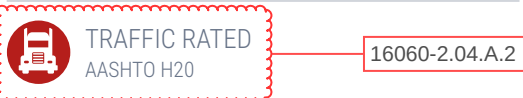
Material:	Reinforced Concrete with Steel Frame
Model:	20" Dia.
Weight:	106 lbs
Wall Type:	Straight
Mouseholes:	0
Performance:	H20, AASHTO



GRADE RING

Material:	Cast Iron
Grade Options:	1" Height 1-1/2" Height 2" Height

LOAD RATING / NOTES



Actual load rating is determined by the box and cover combination.

WARNING: Cancer and Reproductive Harm - www.P65Warnings.ca.gov

All information contained on this sheet is current at the time of publication.
Oldcastle reserves the right to discontinue or update product information without notice.

Revision 05/05/21
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6.0 BONDING JUMPERS

SPECIFICATION – 16060-2.03.E GROUNDING AND BONDING



TYPE B

1-Hole Ferrule End Flexible Copper
Braid, Bonding Jumpers,
Bonding Straps

Flexible copper braid jumpers designed to take up linear expansion and contraction, compensate for misalignment and absorb vibratory movement of electrical equipment and devices. Made of flat extra flexible tinned pure copper braid with high quality BURNDY ferrules on each end. Other lengths, plating and connector sizes are available; contact BURNDY for more information.



E-61

Catalog Number	AWG Equivalent	Length (inches)	End Type	Stud Hole	Approximate Ampere Rating	
					Indoor	Outdoor
BB024L6T14*	#6 AWG	6.00	Ferrule	1/4	75	100
BB024L9T14*	#6 AWG	9.00	Ferrule	1/4	75	100
BB036L9T14*	#5 AWG	9.00	Ferrule	1/4	85	125
BB036L12T14*	#5 AWG	12.00	Ferrule	1/4	85	125
BB036L18T14*	#5 AWG	18.00	Ferrule	1/4	85	125
BB048L6T14*	#4 AWG	6.00	Ferrule	1/4	95	150
BB048L9T14*	#4 AWG	9.00	Ferrule	1/4	95	150
BB067L6T14	#2 AWG	6.00	Ferrule	1/4	110	180
BB067L9T14	#2 AWG	9.00	Ferrule	1/4	110	180
BD6T14	#1 AWG	6.00	Ferrule	1/4	190	225
BD9T14	#1 AWG	9.00	Ferrule	1/4	190	225
BE12T716	3/0 AWG	12.00	Ferrule	7/16	340	405
BE18T716	3/0 AWG	18.00	Ferrule	7/16	340	405
BE12T58	3/0 AWG	12.00	Ferrule	5/8	340	405
BE18T58	3/0 AWG	18.00	Ferrule	5/8	340	405
BE24T58	3/0 AWG	24.00	Ferrule	5/8	340	405
BE6T716	3/0 AWG	6.00	Ferrule	7/16	340	405
BF6T716	4/0 AWG	6.00	Ferrule	7/16	360	430
BF12T716	4/0 AWG	12.00	Ferrule	7/16	360	430
BF18T716	4/0 AWG	18.00	Ferrule	7/16	360	430
BG6T716	300 kcmil	6.00	Ferrule	7/16	415	495
BG8T716	300 kcmil	8.00	Ferrule	7/16	415	495
BG12T716	300 kcmil	12.00	Ferrule	7/16	415	495
BG12T12	300 kcmil	12.00	Ferrule	1/2	415	495

* Not CSA Certified

TYPE B

Flexible Copper Braid

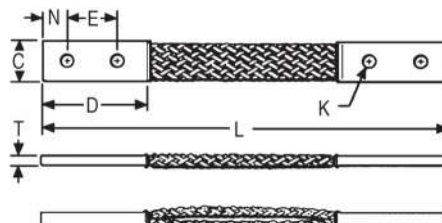
Flexible copper braid jumpers designed to take up linear expansion and contraction, compensate for misalignment and absorb vibratory movement of electrical equipment and devices.

Made of flat extra flexible, tinned, pure copper braid, with unplated, seamless, pure copper ferrules formed into a rectangular shape on each end.



Last two numbers in catalog number indicate total length of braid in inches (e.g., BD12N or BD12 is 12" long braid jumper).

Other lengths, plating and drilling are available. Contact the factory.



Catalog # Prefix	Equiv. AWG Size
BD	#1
BE	3/0
BF	4/0
BG	300 kcmil

Catalog Number	Number of Braids in Ferrules	C	D	E	K	L	N	T	Approximate Ampere Rating	
									Indoor	Outdoor
BD12 ②	1	0.94	2.50	1.25	0.44	12	0.62	0.13	190	225
BD12N ②	1	0.94	3.00	1.75	0.56	12	0.62	0.13	190	225
BD18 ②	1	0.94	2.50	1.25	0.44	18	0.62	0.13	190	225
BD18N ②	1	0.94	3.00	1.75	0.56	18	0.62	0.13	190	225
BD24 ②	1	0.94	2.50	1.25	0.44	24	0.62	0.13	190	225
BD24N ②	1	0.94	3.00	1.75	0.56	24	0.62	0.13	190	225
BE12 ②	1	1.50	3.00	1.50	0.44	12	0.75	0.17	340	405
BE12N ②	1	1.50	3.00	1.75	0.56	12	0.62	0.17	340	405
BE18 ②	1	1.50	3.00	1.50	0.44	18	0.75	0.17	340	405
BE18N ②	1	1.50	3.00	1.75	0.56	18	0.62	0.17	340	405
BE24 ②	1	1.50	3.00	1.50	0.44	24	0.75	0.25	340	405
BE24N ②	1	1.50	3.00	1.75	0.56	24	0.62	0.17	340	405
BF12 ②	1	1.19	3.00	1.50	0.44	12	0.75	0.25	360	430
BF12N ②	1	1.19	3.00	1.75	0.55	12	0.62	0.25	360	430
BF18 ②	1	1.19	3.00	1.50	0.44	18	0.75	0.25	360	430
BF18N ②	1	1.19	3.00	1.50	0.44	18	0.75	0.25	360	430
BF24 ②	1	1.19	3.00	1.50	0.44	24	0.75	0.25	360	430
BF24N ②	1	1.19	3.00	1.75	0.56	24	0.62	0.25	360	430
BG12	1	1.50	3.00	1.50	0.44	12	0.75	0.25	415	495
BG12N ①	1	1.50	3.00	1.75	0.56	12	0.62	0.25	415	495
BG18	1	1.50	3.00	1.50	0.44	18	0.75	0.25	415	495
BG18N ①	1	1.50	3.00	1.75	0.56	18	0.62	0.25	415	495
BG24	1	1.50	3.00	1.50	0.44	24	0.75	0.25	415	495
BG24N ①	1	1.50	3.00	1.75	0.56	24	0.62	0.25	415	495

① Tongue drilled per (2) hole NEMA Standard

② Certified to CSA C22.2, No. 41 Grounding and Bonding Equipment Standards in addition to the UL467 Listing which all items above are Listed to.

NOTE: Equivalent sizes may be designated by suffix letters representing variations in length, mounting configurations, pad size and finish. Contact factory for details.
For Tin plated ferrules add suffix -TN to the catalog number.

TYPES CCY, B-B

Covered Jumpers

Insulated flexible copper braid jumpers designed to take up linear expansion and contraction, compensate for misalignment and absorb vibratory movement of electrical equipment and devices. Made of extra flexible tinned pure copper braid with high quality BURNDY tin plated connectors or ferrules on each end. Other lengths, plating, insulation colors and connector sizes are available; contact BURNDY for more information.



E-65

Catalog Number	AWG Equivalent	Length (inches)	End Type	Stud Hole	Heat Shrink Color
CCY106LT12G	#10 AWG	6.00	Connector	1/2	Green
CCY10L9T12G		9.00		1/2	
CCY10L12T12G		12.00		1/2	
CCY10L18T12G		18.00		1/2	
CCY10L24T12G		24.00		1/2	
CCY10L12LT1090G		12.00		#8-#10	
CCY10L18LT1090G		18.00		#8-#10	
CCY10L24LT1090G		24.00		#8-#10	
CCY10L6LT38G		6.00		3/8	
CCY10L9LT38G		9.00		3/8	
CCY10L12LT38G		12.00		3/8	
CCY10L18LT38G		18.00		3/8	
CCY10L24LT38G		24.00		3/8	
CCY10L7T14GY	#10 AWG	7.00	Connector	1/4	Green & Yellow
CCY10L9T14GY		9.00		1/4	
CCY10L12LT14GY		12.00		1/4	
CCY10L14LT14GY		14.00		1/4	
CCY10L18LT14GY		18.00		1/4	
BB024L12LT14B	#6 AWG	12.00	Connector	1/4	Black
BB024L18LT14B		18.00		1/4	
BB024L24LT14B		24.00		1/4	
BD12NB	#1 AWG	12.00	Ferrule	2 hole NEMA	
BD18NB		18.00		2 hole NEMA	
BD24NB		24.00		2 hole NEMA	

